A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex–Pseudoconvex Constraints

Nguyen Van The[a], Tran Ngoc Thang[a] and Nguyen Hong Son[a]

[a]Faculty of Mathematics and Informatics, Hanoi University of Science and Technology, No. 1 Dai Co Viet Street, Bach Mai Ward, Hanoi 100000, Vietnam

ARTICLE HISTORY
Compiled October 29, 2026

ABSTRACT
We study a nonlinear split feasibility problem whose output set is a normal box and whose component maps split into convex and pseudoconvex blocks. Classical *CQ* solvers presuppose a linear operator, while existing self-adaptive methods require the aggregate proximity function to be convex, which a single pseudoconvex component invalidates. A Tchebyshev scalarization turns feasibility into a closed-form max-type violation function. We minimize it with the Dual-Track Adaptive Projection (DTAP) algorithm, which takes a residual-adaptive step on the convex constraints and a safeguarded diminishing step on the pseudoconvex ones, and projects each iterate onto the base set. Its step sizes need no operator-norm, Lipschitz, or Hölder modulus, yet we prove strong convergence to a feasible point. As an application, we recast two disjoint feature-selection models, one fractional-programming and one fuzzy-rough, as one nonlinear SFP, with no artificial terms; a sparse variant DTAP-S adds a feature budget. On eight UCI benchmarks DTAP matches or improves accuracy and AUC over the base models while reaching feasibility in fewer iterations, and DTAP-S yields compact, stable selections.

KEYWORDS
Split feasibility problem; Pseudoconvex optimization; Tchebyshev scalarization; Adaptive (Lipschitz-free) step size; Strong convergence; Feature selection

AMS CLASSIFICATION
47J25; 90C25; 90C26; 65K05; 68T05

1. Introduction

The *split feasibility problem* (SFP), introduced by Censor and Elfving [6], seeks a point $x^\star \in C \subseteq \mathbb{R}^n$ whose image under a bounded linear operator $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ lies in a convex set $Q \subseteq \mathbb{R}^m$:

$$\text{find } x^\star \in C \quad \text{such that} \quad \mathcal{A}x^\star \in Q. \tag{1}$$

It is a general formulation that specializes to convex feasibility, linear inverse problems, image reconstruction [5], signal processing [9], and radiotherapy planning [7]. Its

Email: the.nv227153@sis.hust.edu.vn
Email: thang.tranngoc@hust.edu.vn
CONTACT Nguyen Hong Son. Email: Son.NH240066D@sis.hust.edu.vn

canonical solver is Byrne's CQ method [5],

$$x^{k+1} = P_C\Big(x^k - \tau\, \mathcal{A}^\top (I - P_Q)\mathcal{A}x^k\Big), \qquad \tau \in \big(0,\, 2/\left\|\mathcal{A}\right\|^2\big). \tag{2}$$

This solver has two drawbacks, namely a step size depending on the operator norm $\|\mathcal{A}\|$ and a projection P_Q with no closed form for generic Q. The second is met by relaxed-projection variants [15, 35], which replace Q by an outer half-space, and the first by the self-adaptive rule of López et al. [25], which calibrates the step from the current value of the merit function $f(x) := \frac{1}{2}\left\|(I - P_Q)\mathcal{A}x\right\|^2$,

$$\tau_n = \frac{\rho_n\, f(x_n)}{\left\|\nabla f(x_n)\right\|^2}, \qquad 0 < \rho_n < 4. \tag{3}$$

Rule (3) is the reference point of this paper, so we isolate the mechanism behind it, beyond its removal of $\|\mathcal{A}\|$. Its analysis turns on the single inequality $\langle \nabla f(x),\, x - z\rangle \geq 2f(x)$, valid at every solution z because $I - P_Q$ is firmly nonexpansive; this inequality delivers both the admissible range $\rho_n \in (0, 4)$ and the per-step decrease $\left\|x_{n+1} - z\right\|^2 \leq \left\|x_n - z\right\|^2 - \rho_n(4 - \rho_n) f^2(x_n)/\left\|\nabla f(x_n)\right\|^2$. Strong convergence is available in Hilbert space through regularized and Krasnosel'skiĭ–Mann variants [34]. The mechanism above is therefore a property of the *composition* $(I - P_Q)\mathcal{A}$: it comes from the linearity of $\mathcal{A}$, and the residual calibration alone does not provide it.

Linearity fails in many contemporary problems, among them the dose maps of intensity-modulated radiotherapy, where the differing densities of organs and tissues make the beam-to-dose relation nonlinear [23], economic equilibrium models [36], and the training of neural predictors [31]. This leads to the *nonlinear split feasibility problem* (NSFP): given closed convex $K \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ and a nonlinear map $F : \mathbb{R}^n \to \mathbb{R}^m$,

$$\text{find } x^\star \in K \quad \text{such that} \quad F(x^\star) \in Q. \tag{4}$$

The NSFP is not itself new. Li et al. [23] treat its multiple-sets form by minimizing a smooth Euclidean proximity function with a self-adaptive, backtracked step, while Yen et al. [36] minimize the max-type merit $\max_i |(I - P_{Q_i})F_i(x)|^2$ over the solution set of an equilibrium problem, driving it by the Clarke subdifferential on the active index set, which is the same scalarization and active-set device we use below. Both papers replace, by a structural hypothesis, the quantitative inequality that linearity used to supply for free: [23] require the aggregate proximity function to be convex, so that its gradient is co-coercive, and [36] require every component map to be *quasilinear*, that is, both quasiconvex and quasiconcave. Neither hypothesis is ours, and neither of these classes coincides with the one we assume. Global convexity of the proximity function fails the moment a single component is only pseudoconvex; two-sided quasilinearity already excludes an ordinary convex map such as $x \mapsto x^2$, which is quasiconvex but not quasiconcave. We ask instead that each component be convex or pseudoconvex, a one-sided condition that these two classes neither contain nor are contained in, as Section 2.2 makes precise. Two further difficulties attend the nonlinear case. When some F_i is nonconvex the global gradient inequality is lost, so CQ-type separating hyperplanes may cut off the solution; and the available step sizes are tied either to a Lipschitz constant of ∇F or, for the wider quasi-convex class, to a Hölder modulus [19], both hard to estimate at scale. Section 3.4 makes these challenges precise as (D1)–(D4).

A complementary motivation comes from *feature selection* [16, 30], where embedded models cast selection as a constrained continuous program over a feature-weight vector. Two validated but isolated models live here: the fractional-programming model of Wang et al. [32], which minimizes a redundancy-to-relevance ratio and is solved by a projection neural network, and the fuzzy-rough model AFRSFCV [33], which fuses class variance into fuzzy lower approximations and is solved by a softmax parametrization. Optimizing either in isolation tends to yield overly sparse selections, and neither offers a common language for combining criteria.

These two lines of work, nonlinear split feasibility on the one hand and embedded feature selection on the other, turn out to be two instances of a single problem. This paper addresses three questions.

(Q1) Is there a structured but broadly applicable subclass of NSFP for which one can design an iterative method that is simultaneously (i) free of operator-norm, Lipschitz, and Hölder constants; (ii) free of projections onto the intractable solution set; and (iii) provably *strongly* convergent?

(Q2) Can heterogeneous selection criteria, some convex and some only pseudoconvex, be expressed in one feasibility model *without inventing new penalty terms*, so that the model inherits the empirical meaning of its constituents?

(Q3) Does solving feature selection *as* a feasibility problem offer any concrete advantage over solving each constituent model on its own, whether in selection quality, stability, or computational cost?

We answer all three questions affirmatively. On the theoretical side we isolate the subclass of NSFP in which the output set is a *normal box* $Q = (-\infty, b_1] \times \cdots \times (-\infty, b_m]$ and the component maps split into a convex group I^{cvx} and a pseudoconvex group I^{ps}. The box structure admits a *Tchebyshev scalarization*: measuring the residual in the sup-norm collapses $d_\infty(F(x), Q)$ into the closed-form max-type function $J(x) = \max_i (F_i(x) - b_i)^+$, which certifies feasibility one constraint at a time and whose maximizers expose the binding constraints. The split into convex and pseudoconvex groups then leads to a *dual-track* step-size policy that reads exactly this information, described next.

We work in finite-dimensional Euclidean space with smooth component maps and a nonempty feasible set; these mild conditions are made precise in Assumption 1. Our target is *feasibility*, that is, returning a point of Ω, rather than the minimization of a designated objective. Any optimization criterion of interest enters the model as one of the constraint functions F_i, so that 'solving' the problem means bringing all criteria below their thresholds. This feasibility viewpoint allows several heterogeneous criteria, convex and pseudoconvex alike, to coexist in a single problem, and it lets a single algorithm with a single parameter setting serve the whole study. The price is that we do not claim optimality of the returned point for any one criterion; we return a point that is acceptable for all of them, which is the requirement in the feature-selection application.

This paper makes five principal contributions.

C1. A tractable nonlinear split-feasibility model. We formulate the *normal-box* NSFP with mixed convex–pseudoconvex components and derive, from a Tchebyshev scalarization, a closed-form max-type violation function J whose zero set coincides with the feasible set. This turns an abstract feasibility requirement into a single nonsmooth merit function whose maximizers identify the binding constraints, which is the structure the dual-track step exploits (Section 4).

C2. The dtap algorithm with Lipschitz-free adaptive steps. We propose the Dual-Track Adaptive Projection algorithm, which decomposes the constraint index set into a convex track and a pseudoconvex track and assigns each the step rule that fits its geometry. On the convex track we extend rule (3) to the nonlinear setting: the step is $\lambda\,\Phi_k^{\mathrm{cvx}}/\left\|g_k^{\mathrm{cvx}}\right\|^2$ with $\lambda = 2\rho_{\mathrm{cvx}} \in (0,4)$, where Φ^{cvx} is a partial quadratic violation and g^{cvx} its gradient. Equivalently it is a Polyak step [29] for Φ^{cvx} with known optimal value zero, but with the *doubled* relaxation range of López et al. rather than the classical $(0,2)$; the doubling is no longer licensed by firm nonexpansiveness, which linearity would have supplied; it now rests on the aggregated inequality $\left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle \geq 2\Phi_k^{\mathrm{cvx}}$ that convexity of the individual F_i still yields. On the pseudoconvex track, where only sign information remains, we use a *safeguarded diminishing step*. The two directions are merged by a soft proportional weight, and each iteration ends with a single projection onto the base set K. No operator norm, Lipschitz, or Hölder constant is needed, and the method projects only onto the base set, never onto the output set Q or the intractable solution set (Section 4).

C3. A strong-convergence theorem. We prove that the DTAP iterates converge *strongly* to a feasible point under three mild assumptions (Theorem 4.12). The proof rests on a novel one-step quasi-Fejér inequality that yields boundedness, distance convergence, and decay of the convex-track violation; feasibility on the pseudoconvex track is then secured by a contradiction argument driven by continuity and pseudoconvexity. The purely convex and purely pseudoconvex regimes follow as corollaries, and a best-violation rate of $O(N^{-1/3})$ is obtained (Section 4).

C4. A unified, constraint-faithful feature-selection framework. We show that the fractional-programming model [32] and the fuzzy-rough-set model AFRS-FCV [33] reduce to a *single* normal-box NSFP whose three constraint functions are taken *unchanged* from the two models' own objectives, namely two convex terms from the fuzzy-rough model and one pseudoconvex ratio from the fractional model. No artificial penalty is introduced, so the unified model preserves the empirical semantics of its constituents while becoming directly solvable by DTAP; new criteria can be appended simply as further convex or pseudoconvex constraints, and a sparse-projection variant DTAP-S adds an explicit, tunable feature budget (Section 5).

C5. Empirical validation. On eight UCI benchmarks spanning a wide range of sample sizes, dimensions, class counts, and balance levels, and against four baselines (the two base models and two subgradient variants for quasi-convex feasibility [19]), DTAP preserves or improves accuracy and AUC over the base models, with the gains concentrated on the SVM classifier and significant at the nominal level under Wilcoxon and Friedman tests, and it produces stable, consistent feature sets. Against the same-class subgradient methods DTAP yields nearly identical selections yet reaches feasibility in substantially fewer iterations and lower wall-clock time, up to three orders of magnitude on the larger datasets, while avoiding the Hölder-modulus estimation those methods require (Section 6).

The paper is organized as follows. Section 2 reviews the literature and positions DTAP against twelve representative prior works. Section 3 collects the analytical preliminaries. Section 4 states the normal-box NSFP, derives the Tchebyshev merit function, presents the dual-track decomposition and the standing assumptions, specifies DTAP, and then develops the convergence analysis. Section 5 builds the unified feature-

selection framework. Section 6 reports the numerical study, and Section 7 concludes with limitations and extensions.

2. Related work

We review four threads (linear split feasibility, nonlinear and quasi-convex feasibility, optimization-based feature selection, and decomposition) and position DTAP in Table 1.

2.1. Linear split feasibility

Introduced by Censor and Elfving [6], the SFP was simplified by Byrne's CQ method [5], which uses only P_C, P_Q and a gradient step through $\mathcal{A}^\top(I - P_Q)\mathcal{A}$ but pays with a step size tied to $\|\mathcal{A}\|^2$ and a projection P_Q explicit only for structured Q. Two lines address these costs: relaxed-projection methods [15, 35] replace Q by an outer half-space with closed-form projection, and the self-adaptive rule (3) of López et al. [25] calibrates the step from the current residual, removing $\|\mathcal{A}\|$. Xu [34] proved *strong* convergence for regularized and Krasnosel'skiĭ–Mann variants in Hilbert space.

Rule (3) has since been extended beyond convex sets. Chen et al. [8] keep it verbatim, as $\rho_k = \lambda_k f_k(u_k)/\left\|\nabla f_k(u_k)\right\|^2$ over relaxed level sets $C_{k,\xi}, Q_{k,\xi}$, and allow the level-set functions c, q to be continuous but nonconvex by replacing the subdifferential with an *S-subdifferential*: a modulus $r_f > 0$ must exist for which $f + \frac{r_f}{2} d_S^2$ is subdifferentiable, equivalently $\partial_{S,r_f} f(u) = \partial f(u) + r_f(I - P_S)(u)$. This is a genuine relaxation of convexity, but it is neither ours nor comparable to ours: a pseudoconvex F_j need not admit such an r_f, and a function that does need not be pseudoconvex. What matters more for the present paper is where the nonconvexity is placed. In [8] the operator remains a *matrix*, so the merit function $f_k(u) = \frac{1}{2}\left\|\mathcal{A}u - P_{Q_{k,\xi}}\mathcal{A}u\right\|^2$ is a convex composition and the subgradient inequality that drives their proof is available for free; the nonconvexity sits in the sets, never in the map. Their relaxation range is accordingly $\lambda_k \in (0, 2)$, obtained from the plain gradient inequality rather than from the co-coercivity that yields López et al.'s $(0, 4)$. The residual-based, constant-free philosophy of this thread and its strong-convergence target inform our design; the mechanism that licenses it does not carry over to a nonlinear F, and Section 2.2 takes up what replaces it.

2.2. Nonlinear and quasi-convex feasibility

When the map is nonlinear, the global gradient inequality fails and CQ-type hyperplanes may exclude the solution. The closest predecessor is Li et al. [23], who pose the nonlinear *multiple-sets* SFP (find $x \in \bigcap_i C_i$ with $F(x) \in \bigcap_j Q_j$ for a continuously differentiable F) and minimize over a bounded convex $\Omega \supseteq \bigcap_i C_i$ the Euclidean proximity function $p(x) = \frac{1}{2}\sum_i \alpha_i \left\|(I - P_{C_i})x\right\|^2 + \frac{1}{2}\sum_j \beta_j \left\|(I - P_{Q_j})F(x)\right\|^2$ by the projected step $x^{k+1} = P_\Omega[x^k - \beta_{k+1}\nabla p(x^k)]$, the step β_{k+1} being chosen self-adaptively by backtracking. That framework does not sit beside our geometry; it contains it. A normal box is the intersection $\bigcap_{i=1}^m \{z \in \mathbb{R}^m : z_i \le b_i\}$ of m half-spaces, so with $C_1 = K$ and the componentwise violation $v_i(x) := (F_i(x) - b_i)^+$ one has $\left\|(I - P_{Q_i})F(x)\right\| = v_i(x)$

and therefore

$$p(x) = \tfrac{1}{2}\sum_{i=1}^{m} \beta_i\, v_i(x)^2, \qquad \nabla p(x) = \sum_{i=1}^{m} \beta_i\, v_i(x)\, \nabla F_i(x). \tag{5}$$

Comparing (5) with the DTAP direction (16) below, g^{cvx} is exactly ∇p with unit weights, restricted to the δ-active convex indices. The direction is therefore not where the novelty lies, and we do not claim it there; the novelty lies in what makes the step legitimate. The convergence proof of [23] runs entirely on co-coercivity of ∇p, which their Lemma 2.4 obtains from *convexity of* p together with a Lipschitz ∇p on the bounded Ω; co-coercivity also keeps the right-hand side of their backtracking test nonnegative, and hence makes the search terminate at all. On a normal box, p is convex precisely when every F_i is convex, since then $v_i = (F_i - b_i)^+$ is convex and nonnegative and $t \mapsto t^2$ is convex and increasing on $[0, \infty)$. A single pseudoconvex component invalidates these implications: v_j^2 need not be convex, p need not be convex, and ∇p need not be monotone, so the right-hand side of their backtracking test may become negative.[1] When it does, no positive step size satisfies the test, and the line search is not well-defined. DTAP replaces the global and unverifiable hypothesis 'p is convex' by a classification that can be checked one constraint at a time, and replaces the search by a closed-form step; it also drops the requirements that Ω be bounded and that $F, \nabla F$ be globally Lipschitz, deriving boundedness of the iterates instead (Lemma 4.9).

The other close nonlinear predecessor is Yen et al. [36], and the overlap is at the level of the merit function and its differentiation, not merely the topic. On a normal box their componentwise term $p_i(x) = |(I - P_{Q_i})F_i(x)|^2$ equals $v_i(x)^2$ and their gradient $\nabla p_i(x) = 2(I - P_{Q_i})F_i(x)\nabla F_i(x)$ equals $2v_i(x)\nabla F_i(x)$, so their scalarized objective $\max_i p_i$ is the square of our J and their search direction, a convex combination of active ∇p_i selected through the active set $I(x) = \{i : p_i(x) = p(x)\}$ and the Clarke identity $\partial^0 p(x) = \mathrm{co}\{\nabla p_i(x) : i \in I(x)\}$, is exactly the mechanism our tracks (16)–(17) implement. We do not reintroduce that mechanism as novel. Three differences separate the methods, and each is verifiable. First, the admissible map class. Yen et al. require every F_i to be *quasilinear*; we require convex or pseudoconvex. Neither class contains the other: $x \mapsto x^2$ is convex, hence covered here, but not quasiconcave, hence excluded by their (A2); conversely a monotone cubic $x \mapsto x^3$ is quasilinear, hence covered by them, but has a stationary point that is not a minimizer and so is neither convex nor pseudoconvex. Second, the driving step. Their feasibility direction is a normalized subgradient with a diminishing step $\alpha_k = \beta_k / \max\{\delta_k, \|g_k\|\}$ under $\sum \beta_k^2 < \infty$; on our convex track we instead use the residual-adaptive rule (3), whose length is set by the current violation rather than by a prescribed sequence, and which is not diminishing. Third, the source set. Their C is the solution set of a paramonotone equilibrium problem, handled by a second projected-subgradient track, whereas ours is the explicit base set K; this is a deliberate modelling choice, since in our target application K is a simplex or box onto which projection is closed-form, and it lets the second track address the *pseudoconvex output constraints* rather than an implicit source constraint. In short, Yen et al. decompose source-versus-image; DTAP decomposes the image constraints by their individual geometry.

The remaining nonlinear thread is quasi-convex feasibility. Tuan et al. [31] embed

[1] In the notation of Eq. (3.3) of [23], the acceptance test requires $\beta_{k+1}\left\|\nabla p(x^k) - \nabla p(x^{k+1})\right\|^2 \le (2 - \mu)\left\langle x^k - x^{k+1}, \nabla p(x^k) - \nabla p(x^{k+1})\right\rangle$; its right-hand side is nonnegative for every pair of iterates iff ∇p is monotone, which on a normal box holds iff every F_i is convex.

split-feasibility constraints in controllable Pareto-front learning, and Hu et al. [19] solve systems of quasi-convex inequalities by subgradient methods with constant and dynamic steps. Their constraint class is strictly *larger* than ours: convex and pseudoconvex functions are quasi-convex, so (8) is one of their instances, and we accordingly use their two variants as baselines in Section 6. This is the opposite relation to the one with Yen et al.: Hu et al. sit outside and above our class, Yen et al. beside it. Generality has a price. A uniform treatment of the whole quasi-convex class leaves only sign information on every constraint, so their dynamic step must be calibrated by a Hölder modulus that is large and unreliable on real data. DTAP does not enlarge the class; it exploits the finer convex/pseudoconvex split *inside* it to recover a quantitative gradient inequality on part of the system, and with it a step that needs no modulus at all.

2.3. Optimization-based feature selection

Two embedded, optimization-based models anchor our application (see the surveys [16, 24, 30]). Wang et al. [32] pose selection as a fractional program (a redundancy quadratic over a relevance linear form on the simplex, pseudoconvex there), solved by a projection neural network. Wang et al. [33] propose AFRSFCV, a fuzzy-rough model injecting class variance into the similarity relation and minimizing a softmax loss that combines a lower-approximation fit, a decision-consistency term, and Tikhonov regularization. Each ships its own solver and objective; neither combines criteria, and optimizing either alone tends to yield overly sparse selections. Our framework reuses exactly these objectives as feasibility constraints, retaining their semantics while unifying them.

2.4. Decomposition and the positioning of **dtap**

Decomposing a heterogeneous problem into homogeneous blocks updated by block-appropriate rules is a recurring theme [3, 9]. DTAP applies this at the level of *constraints*: a residual-adaptive step where a quantitative gradient inequality holds (convex track), and a safeguarded step where only sign information is available (pseudoconvex track). A third strand pursues strong convergence via Halpern, viscosity, or inertial modifications [9, 34], but typically for linear SFP or a single averaged operator; DTAP reaches it by constructing a tailored one-step quasi-Fejér inequality (following the general framework of Combettes 10) plus a pseudoconvex sign argument, needing neither the exact Lipschitz constant nor the projection onto the complex feasible set.

Table 1 makes the positioning explicit: no single column of it isolates DTAP, because each ingredient of the method already has precedent. A constant-free step appears in [25] and, in diminishing form, in [36]; the same residual-adaptive formula is reused verbatim by [8], though that method additionally needs the S-subdifferential modulus and so is not itself constant-free. Projecting only onto the base set is done by both [19] and [36]; whole-sequence convergence is proved in [23, 36]; and a max-type scalarization driven by the Clarke subdifferential on an active set is the device of [36]. The contribution of DTAP lies in the conjunction of two elements not previously combined: the admissible *class* and the *step that drives it*. On the class, the two nearest neighbours sit on opposite sides: [19] lies strictly above ours (quasi-convex), while [36] lies beside it (quasilinear, which neither contains nor is contained in convex-plus-pseudoconvex). Consequently neither subsumes the problem we solve. On the step, the residual-adaptive rule (3) has always been licensed by a quantitative inequality

Table 1. Positioning of DTAP against twelve representative prior works. Two columns carry ✓/×: 'Const.-free' marks methods whose iteration needs no prescribed constant: no operator norm, no Lipschitz or Hölder modulus, and no S-subdifferential modulus; 'Only P_K' marks methods that project only onto the base set, with no projection onto the output set or the solution set. 'Step-size rule' gives the update in closed form, and P_Ω in it denotes a projection onto the base restriction set of the multiple-sets model. No single column isolates DTAP: the 'Const.-free' tick is shared with [23, 25, 35, 36] and the 'Only P_K' tick with [19, 32, 36], so in particular Yen et al. carry both. DTAP is set apart only by the conjunction of the last two columns, specifically a constant-free, residual-adaptive step on a *mixed convex–pseudoconvex* class. On that class the two nearest neighbours lie on opposite sides: the quasi-convex row of [19] is strictly *broader* (it contains convex-plus-pseudoconvex, but treats it uniformly and pays with a Hölder modulus), whereas the quasilinear row of [36] is *incomparable* (it neither contains nor is contained in our class), so neither subsumes the problem DTAP solves.

Work	**Problem class**	**Map**	**Output set**	**Step-size rule**	**Const.-free**	**Only P_K**	**Constraint**
Censor–Elfving [6]	Split feas. (origin)	Lin.	General convex	Bregman multiprojection	×	×	convex
Byrne [5] (CQ)	Split feasibility	Lin.	General convex	$\tau < 2/\|\mathcal{A}\|^2$ fixed	×	×	convex
Fukushima [15]	Variational inequality	Lin.	Level set of cvx c	Relaxed (half-space)	×	×	convex
Yang [35] (relaxed CQ)	Split feasibility	Lin.	Level set of cvx c	$\rho_n f/\|\nabla f\|^2$, half-sp.	✓	×	convex
López et al. [25]	Split feasibility	Lin.	General convex	$\rho_n f/\|\nabla f\|^2$, $\rho_n \in (0,4)$	✓	×	convex
Xu [34]	Split feas. (Hilbert)	Lin.	General convex	KM / regularized	×	×	convex
Li–Han–Zhang [23]	Nonlinear MSFP	Nonlin.	$\bigcap_j$ convex	β_k backtracked (needs p cvx)	✓	×	convex (p cvx)
Yen et al. [36]	Nonlin. SFP (Nash)	Nonlin.	Convex	$\beta_k/\max\{\delta_k, \|g_k\|\}$ dimin.	✓	✓	quasilinear
Chen et al. [8]	Nonconvex SFP	Lin.	S-subdiff. level sets	$\lambda_k f_k/\|\nabla f_k\|^2$, $\lambda_k \in (0,2)$	×	×	S-subdiff. noncvx
Hu et al. [19]	Quasicvx feasibility	Nonlin.	Inequality system	Const./dynamic subgrad. (Hölder)	×	✓	quasi-cvx
Wang et al. [32]	Frac.-prog. selection	Nonlin.	Simplex	Neurodynamic (PNN)	×	✓	pseudo-cvx
Wang et al. [33]	Fuzzy-rough selection	Nonlin.	$\mathbb{R}^m$ (softmax)	Gradient descent	×	×	nonconvex
This work	**Normal-box NSFP**	**Nonlin.**	**Normal box**	$\lambda\Phi^{\mathrm{cvx}}/\|g^{\mathrm{cvx}}\|^2 \oplus \gamma_k$	✓	✓	**cvx + pseudo**

that its setting supplies for free: firm nonexpansiveness of $I - P_Q$ under a linear $\mathcal{A}$ in [8, 25], and co-coercivity of ∇p under a convex proximity function in [23]. We obtain the same inequality, $\langle g^{\mathrm{cvx}}, x - x^\star \rangle \geq 2\Phi^{\mathrm{cvx}}$, from convexity of the individual components, which persists under both a nonlinear F and the presence of pseudoconvex components elsewhere in the system; those components are then absorbed on a second track that uses only the sign pseudoconvexity still guarantees. This inequality lets the convex part keep a non-diminishing, residual-scaled step, whereas [36] must fall back on a prescribed diminishing sequence. To our knowledge DTAP is therefore the first method for the normal-box NSFP with *mixed convex–pseudoconvex* components whose step is closed-form and free of every constant (operator norm, Lipschitz constant, Hölder modulus, S-subdifferential modulus), which requires neither linearity of the map nor convexity of the aggregate proximity function nor two-sided quasilinearity of the components, and which converges strongly to a feasible point.

3. Preliminaries

Throughout, $H := \mathbb{R}^n$ denotes the n-dimensional Euclidean space with inner product $\langle x, y \rangle := \sum_{i=1}^n x_i y_i$ and induced norm $\|x\| := \sqrt{\langle x, x \rangle}$. Every bounded subset of H is relatively compact, so by Bolzano–Weierstrass every bounded sequence has a convergent subsequence; convergence is in norm, $x^k \to x$ iff $\|x^k - x\| \to 0$. Most statements below are standard [1, 2, 27]; the two results invoked directly in the convergence analysis (nonexpansiveness of the projection and a deterministic quasi-Fejér lemma) are proved in Section 4.

We collect the main notation here. The ambient space is $H = \mathbb{R}^n$ with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$; K is the base closed convex set (the simplex Δ_{p-1} in the application) and P_K its metric projection. The component map $F = (F_1, \ldots, F_m)$, threshold vector b, and normal box $Q = \prod_i (-\infty, b_i]$ define the feasible set $\Omega = \{x \in K : F_i(x) \leq b_i \ \forall i\}$; the Tchebyshev error is $J(x) = \max_i (F_i(x) - b_i)^+$ and the componentwise violation $v_i(x) = (F_i(x) - b_i)^+$. The constraint indices split into convex and pseudoconvex sets $I^{\mathrm{cvx}}, I^{\mathrm{ps}}$, with per-track maximal violations $J^{\mathrm{cvx}}, J^{\mathrm{ps}}$, δ-approximate active sets $K^{\mathrm{cvx}}(x), K^{\mathrm{ps}}(x)$, search directions $g^{\mathrm{cvx}}, g^{\mathrm{ps}}$, step sizes $\eta_k^{\mathrm{cvx}}, \eta_k^{\mathrm{ps}}$, soft weight μ_k, and Polyak numerator $\Phi^{\mathrm{cvx}}(x)$; the algorithm parameters are the Polyak relaxation $\rho_{\mathrm{cvx}} \in (0, 2)$, the safeguarded sequence $\{\gamma_k\}$ with $\sum \gamma_k = \infty$ and $\sum \gamma_k^2 < \infty$, and the active-set width $\delta > 0$. In the application, n, p, r are the numbers of samples, features, and classes, $w \in \Delta_{p-1}$ the feature-weight vector, ρ and Q the Fisher-score vector and redundancy matrix, $\underline{R_k}(D), P, Y$ the lower-approximation, decision, and label matrices, and θ the cumulative selection threshold.

3.1. Euclidean projection

Definition 3.1 (Metric projection)**.** Let $K \subseteq H$ be nonempty closed convex. For $x \in H$, a point $p \in K$ with $\|p - x\| = \inf_{y \in K} \|y - x\|$ is the *projection* of x onto K, written $p = P_K(x)$. The map $P_K : H \to K$ is the *metric projection* onto K.

For closed convex K, $P_K(x)$ exists and is unique for every $x \in H$ [2], and is characterized variationally as follows.

Proposition 3.2 (Variational characterization)**.** *Let $K \subseteq H$ be nonempty closed con-*

vex. For $x \in H$ and $p \in K$, one has $p = P_K(x)$ if and only if

$$\langle x - p,\, y - p \rangle \le 0 \qquad \textit{for all } y \in K. \tag{6}$$

Proposition 3.2 underlies the classical regularity properties of the projection: nonexpansiveness, firm nonexpansiveness, and inverse strong monotonicity [2]. Nonexpansiveness is used repeatedly in Section 4 and is isolated there as Lemma 4.2. The simplex projection, needed for the feature-selection application, is computable in $O(m \log m)$ time [11] and is treated as a black box.

3.2. Convex and pseudoconvex functions

Let K be a nonempty convex subset of $H = \mathbb{R}^n$.

Definition 3.3 (Convex function)**.** $f : K \to \mathbb{R}$ is *convex on* K if for all $x, y \in K$ and $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \le \lambda f(x) + (1 - \lambda) f(y)$.

The next notion, introduced by Mangasarian [26], relaxes convexity while preserving the property that a stationary point is a global minimizer.

Definition 3.4 (Pseudoconvex function)**.** Let $f : K \to \mathbb{R}$ be differentiable on an open neighbourhood of K. Then f is *pseudoconvex on* K if for all $x, y \in K$, $\langle \nabla f(x),\, y - x \rangle \ge 0 \Rightarrow f(y) \ge f(x)$.

Definition 3.4 is equivalent to its contrapositive: for all $x, y \in K$,

$$f(y) < f(x) \implies \langle \nabla f(x),\, y - x \rangle < 0. \tag{7}$$

Form (7) is the main tool of the pseudoconvex-track analysis in Section 4: it converts a strict decrease in function value into a strict sign condition on the directional derivative.

3.3. Feature selection

Feature selection is a preprocessing step that compresses an initial feature set into a subset retaining information relevant to the target while discarding redundant and irrelevant features [16, 30].

Definition 3.5 (Feature selection)**.** Let $U = \{x_1, \ldots, x_n\}$ be n samples and $A = \{a_1, \ldots, a_p\}$ a set of p features, with data matrix $X \in \mathbb{R}^{n \times p}$ whose k-th column holds the values of a_k. *Feature selection* seeks $A^\star \subseteq A$ preserving the discriminative information of A for a given learning target while $|A^\star| \ll p$.

Following [30], methods are categorized by label usage (supervised, unsupervised, semi-supervised); by interaction with the learner (filter, wrapper, embedded); and by output form (ranking, subset, weighting). This paper concerns *supervised, embedded, weighting* methods: those that select features by solving a constrained continuous optimization problem whose output is a weight vector $w \in \mathbb{R}^p$ of relative feature importances. This is the class that can be recast as finding a common point of two constraint systems on two spaces (one expressing admissibility of the weight vector,

the other effectiveness of the selected set), which is the structure of the split feasibility problem treated next.

3.4. Split feasibility and its nonlinear extension

Byrne's CQ recursion (2) is Fejér-monotone with respect to the solution set but requires $\|\mathcal{A}\|$ and P_Q. The relaxed methods of [15, 35] remove P_Q by projecting onto an outer half-space, and the self-adaptive rule of [25] removes $\|\mathcal{A}\|$ by calibrating the step from the residual; we inherit the latter, constant-free philosophy. Replacing the linear $\mathcal{A}$ by a nonlinear F yields NSFP (4), which raises four specific challenges that our model and algorithm meet.

(D1) *Loss of the global gradient inequality.* When some F_i are nonconvex, the supporting-hyperplane inequality fails globally, so hyperplanes built from ∇F_i may cut off the solution $x^\star$ and invalidate CQ-type convergence.

(D2) *Choice of the residual on the output set.* For a *normal box* $Q = (-\infty, b_1] \times \cdots \times (-\infty, b_m]$ [18] both the Euclidean and the sup-norm residuals are available in closed form, the former as $d_2(F(x), Q)^2 = \sum_i v_i(x)^2$ and the latter as $d_\infty(F(x), Q) = \max_i v_i(x)$ with $v_i(x) = (F_i(x) - b_i)^+$. The Euclidean choice is the proximity function of the multiple-sets solver [23], which aggregates all violations into one smooth surrogate; that aggregation, however, forces a uniform step on the whole system. The sup-norm residual instead certifies feasibility constraint by constraint ($J(x) = 0$ iff every $v_i(x) = 0$), and its maximizers single out the binding constraints, which is the information a per-constraint step policy requires. We therefore adopt the Tchebyshev scalarization for its structure, which matches the dual-track design of Section 4.2; the choice is not forced by closed-form availability, since the Euclidean residual has one too.

(D3) *Nonsmoothness of the max-type residual.* The Tchebyshev residual is a maximum of componentwise violations, nondifferentiable where several constraints are simultaneously active; the full Clarke subdifferential is delicate when the active set changes between iterations.

(D4) *Dependence on a Lipschitz constant.* Methods for NSFP typically need a step tied to the Lipschitz constant of ∇F, hard to estimate; an adaptive, constant-free step is preferable.

To address (D1)–(D4) together, we add the structural hypothesis that each F_i is convex or pseudoconvex, adopt the constraint-certifying Tchebyshev residual of (D2), treat (D3) through an approximate active set, and design a Lipschitz-free adaptive step for (D4).

4. The dtap algorithm and its convergence analysis

Building on the preliminaries of Section 3, we now formulate the normal-box NSFP and its Tchebyshev scalarization, decompose the constraint system into a convex and a pseudoconvex track, fix the standing assumptions, and state the DTAP algorithm; we then establish its strong convergence to a feasible point.

4.1. Problem statement and Tchebyshev scalarization

Let $H := \mathbb{R}^n$, let $K \subseteq H$ be nonempty closed convex, fix an integer $m \geq 1$, let $F = (F_1, \ldots, F_m) : H \to \mathbb{R}^m$, let $b = (b_1, \ldots, b_m) \in \mathbb{R}^m$, and set the normal box $Q := (-\infty, b_1] \times \cdots \times (-\infty, b_m]$. We study the instance of (4)

$$\text{find } x^\star \in \Omega, \qquad \Omega := \big\{x \in K : F_i(x) \leq b_i,\ i = 1, \ldots, m\big\}, \tag{8}$$

and call Ω the *feasible set*. To turn $F(x) \in Q$ into a tractable merit function, we measure the residual in the sup-norm. For $z = F(x)$, $d_\infty(z, Q) = \inf_{q \leq b} \max_{1 \leq i \leq m} |z_i - q_i|$. By the box structure the infimum separates over coordinates: if $z_i \leq b_i$ then $\inf_{q_i \leq b_i} |z_i - q_i| = 0$ (at $q_i = z_i$), and if $z_i > b_i$ then it equals $z_i - b_i$ (at $q_i = b_i$). Hence $\inf_{q_i \leq b_i} |z_i - q_i| = (z_i - b_i)^+$ with $(t)^+ := \max\{t, 0\}$, so

$$d_\infty(z, Q) = \max_{1 \leq i \leq m} (z_i - b_i)^+. \tag{9}$$

Define the *error function*

$$J(x) := d_\infty\big(F(x), Q\big) = \max_{1 \leq i \leq m} \big(F_i(x) - b_i\big)^+. \tag{10}$$

Then $F(x) \in Q$ iff $J(x) = 0$, so finding a feasible point amounts to minimizing J over K. The nonsmoothness of J (difficulty (D3)) is handled in Section 4.2 by an approximate active set. Formula (10) settles (D2): computed for the price of m scalar evaluations, J vanishes exactly at feasible points and its maximizers pick out the binding constraints, so it is at once a merit function and the selector the dual-track step will read.

4.2. Decomposition and dual-track scalarization

A common strategy in large-scale optimization is not to solve a heterogeneous problem with one uniform technique, but to decompose it into structurally homogeneous blocks and update each with the rule best suited to it [3]. In large-scale sparse regression, for instance, the objective splits into a smooth data-fidelity part and a nonsmooth regularizer, each handled by its own update; the success of such schemes comes from exploiting the structure of each block rather than imposing a common treatment.

Problem (8) has an analogously heterogeneous constraint system: most constraints have simple geometry (convex, even affine), while a few are more complex: nonconvex but pseudoconvex. Folding all violations into one error function J and treating them uniformly would force the safeguarded diminishing step on the whole system, wasting the structure of the convex block and slowing convergence. Following the decomposition philosophy, we split the constraint index set into a *convex track* and a *pseudoconvex track* and give each its own step rule: a Polyak step [29] on the convex track and a safeguarded diminishing step on the pseudoconvex track. The two are reconciled by a soft proportional weight.

Fix a parameter $\delta > 0$. For $x \in H$ define the componentwise violations and the maximal violation

$$v_i(x) := \big(F_i(x) - b_i\big)^+, \qquad J(x) = \max_{1 \leq i \leq m} v_i(x). \tag{11}$$

Thus $x \in \Omega$ iff $x \in K$ and $J(x) = 0$. Partition $I := \{1, \dots, m\}$ into disjoint groups $I = I^{\text{cvx}} \cup I^{\text{ps}}$, where I^{cvx} indexes the convex constraints and I^{ps} the pseudoconvex ones (made precise in Assumption 1). The *per-track maximal violations* are

$$J^{\text{cvx}}(x) := \max_{i \in I^{\text{cvx}}} v_i(x) \quad (=0 \text{ if } I^{\text{cvx}} = \emptyset), \tag{12}$$

$$J^{\text{ps}}(x) := \max_{j \in I^{\text{ps}}} v_j(x) \quad (=0 \text{ if } I^{\text{ps}} = \emptyset), \tag{13}$$

so $J(x) = \max\{J^{\text{cvx}}(x), J^{\text{ps}}(x)\}$. To cope with the nonsmoothness of the maximum (D3), we replace the exact maximizers by a δ-enlarged *approximate active set* on each track:

$$K^{\text{cvx}}(x) := \begin{cases} \{i \in I^{\text{cvx}} : v_i(x) \geq J^{\text{cvx}}(x) - \delta\} & \text{if } J^{\text{cvx}}(x) > 0, \\ \emptyset & \text{if } J^{\text{cvx}}(x) = 0, \end{cases} \tag{14}$$

$$K^{\text{ps}}(x) := \begin{cases} \{j \in I^{\text{ps}} : v_j(x) \geq J^{\text{ps}}(x) - \delta\} & \text{if } J^{\text{ps}}(x) > 0, \\ \emptyset & \text{if } J^{\text{ps}}(x) = 0. \end{cases} \tag{15}$$

On these sets we form the *per-track local functions and directions*

$$\Phi^{\text{cvx}}(x) := \tfrac{1}{2} \sum_{i \in K^{\text{cvx}}(x)} v_i(x)^2, \qquad g^{\text{cvx}}(x) := \sum_{i \in K^{\text{cvx}}(x)} v_i(x) \nabla F_i(x), \tag{16}$$

$$\Phi^{\text{ps}}(x) := \tfrac{1}{2} \sum_{j \in K^{\text{ps}}(x)} v_j(x)^2, \qquad g^{\text{ps}}(x) := \sum_{j \in K^{\text{ps}}(x)} v_j(x) \nabla F_j(x). \tag{17}$$

The vectors $g^{\text{cvx}}(x), g^{\text{ps}}(x)$ are the two search directions, and $\Phi^{\text{cvx}}(x)$ supplies the Polyak numerator on the convex track. Along the iterates we abbreviate $J_k^{\text{cvx}} := J^{\text{cvx}}(x^k)$, $g_k^{\text{cvx}} := g^{\text{cvx}}(x^k)$, $\Phi_k^{\text{cvx}} := \Phi^{\text{cvx}}(x^k)$, and likewise on the pseudoconvex track.

4.3. Standing assumptions

Assumption 1 (Standing assumptions)**.** We assume (A1)–(A3):

(A1) (a) $\Omega \neq \emptyset$. (b) $I = \{1, \dots, m\}$ is partitioned as $I = I^{\text{cvx}} \cup I^{\text{ps}}$; for each i, $F_i : H \to \mathbb{R}$, and the restriction $F_i|_K$ is convex if $i \in I^{\text{cvx}}$ and pseudoconvex if $i \in I^{\text{ps}}$.

(A2) Each $F_i : H \to \mathbb{R}$ is differentiable on H, and ∇F_i is continuous on every compact subset of H.

(A3) (i) $\delta > 0$ fixed. (ii) $\rho_{\text{cvx}} \in (0, 2)$. (iii) $\{\gamma_k\} \subset (0, \infty)$ with $\sum_{k=0}^{\infty} \gamma_k^2 < \infty$ and $\sum_{k=0}^{\infty} \gamma_k = \infty$.

Example 4.1 (A valid step sequence)**.** The sequence $\gamma_k := 1/(k+1)$ satisfies (A3)(iii), since $\sum_{k \geq 0} (k+1)^{-2} < \infty$ while $\sum_{k \geq 0} (k+1)^{-1} = \infty$. More generally $\gamma_k = (k+c)^{-s}$ with $c > 0$ and $s \in (1/2, 1]$ qualifies; the experiments use $s = 0.55$.

4.4. The dtap *algorithm*

The Dual-Track Adaptive Projection algorithm combines the two track-specific step sizes through the soft weight μ_k and projects the combined update onto the base

convex set K. It is stated as Algorithm 1.

Algorithm 1 Dual-Track Adaptive Projection (DTAP) for the normal-box NSFP (8).

Input: map $F = (F_1, \dots, F_m)$, thresholds b, base set K, index split $I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$.
Parameters: $\delta > 0$, tolerance $\varepsilon > 0$, $\rho_{\mathrm{cvx}} \in (0, 2)$, sequence $\{\gamma_k\}$ satisfying (A3)(iii).
Initialization: choose $x^0 \in K$; set $k \leftarrow 0$.

Step 1 (per-track quantities).

(1a) Compute $v_i(x^k)$ for all i, and $J_k^{\mathrm{cvx}} = \max_{i \in I^{\mathrm{cvx}}} v_i(x^k)$, $J_k^{\mathrm{ps}} = \max_{j \in I^{\mathrm{ps}}} v_j(x^k)$ (each 0 if its track is empty).

(1b) Form the approximate active sets $K_k^{\mathrm{cvx}}, K_k^{\mathrm{ps}}$ via (14)–(15).

(1c) Form Φ_k^{cvx}, g_k^{cvx}, g_k^{ps} via (16)–(17).

(1d) Per-track step sizes (Polyak on the convex track, safeguarded on the pseudoconvex track):

$$\eta_k^{\mathrm{cvx}} = \begin{cases} \rho_{\mathrm{cvx}} \dfrac{2\Phi_k^{\mathrm{cvx}}}{\|g_k^{\mathrm{cvx}}\|^2} & \text{if } \|g_k^{\mathrm{cvx}}\| > 0, \\ 0 & \text{otherwise,} \end{cases} \qquad \eta_k^{\mathrm{ps}} = \frac{\gamma_k}{\max\{1, \|g_k^{\mathrm{ps}}\|\}}.$$

(1e) Soft proportional weight: $\mu_k = \dfrac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}}$ if $J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}} > 0$, and $\mu_k = 1/2$ otherwise.

(1f) Combine the two directions: $y^k = x^k - [\mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1 - \mu_k) \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}]$.

Step 2 (projection). $x^{k+1} = P_K(y^k)$.

Step 3 (stopping). If $J(x^k) \le \varepsilon$, stop and return x^k; otherwise set $k \leftarrow k + 1$ and go to Step 1.

For the analysis we write the combined increment as

$$d_k := \mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1 - \mu_k) \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}, \tag{18}$$

so $y^k = x^k - d_k$ and $x^{k+1} = P_K(x^k - d_k)$.

4.5. Per-iteration cost

Each DTAP iteration performs a fixed set of F-agnostic operations: it evaluates the componentwise violations $v_i(x^k)$ and the active-set gradients $\{\nabla F_i(x^k) : i \in K_k^{\mathrm{cvx}} \cup K_k^{\mathrm{ps}}\}$, forms the two search directions $g_k^{\mathrm{cvx}}, g_k^{\mathrm{ps}}$ as weighted sums of these gradients, computes the two scalar step sizes $\eta_k^{\mathrm{cvx}}, \eta_k^{\mathrm{ps}}$, and closes with a single projection P_K onto the base set. Unlike Byrne's CQ recursion (2), there is no power iteration to estimate an operator norm $\|\mathcal{A}\|$ and no line search; the cost is thus dominated by the constraint evaluations and the one projection. The concrete order for the feature-selection instance, which depends on the explicit form of the F_i, is deferred to Section 5 (Remark 4).

The convergence analysis of DTAP is more involved than for classical CQ schemes because the two constraint tracks are asymmetric: on the convex track the gradient inequality gives a quantitative lower bound of order $v_i(x)^2$, hence a strict decrease, whereas on the pseudoconvex track the contrapositive of pseudoconvexity gives only sign information. The two are reconciled in a single one-step quasi-Fejér inequality with a carefully chosen Young parameter, and a contradiction argument rules out cluster points outside the feasible set. We then obtain the main result, strong convergence to a feasible point (Theorem 4.12), and deduce the two degenerate structural regimes as corollaries.

Throughout, $\{x^k\}$ is the infinite sequence generated by Algorithm 1 with the stopping test removed (i.e. $\varepsilon = 0$ in Step 3). As shown below (Lemma 4.10 and Theorem 4.12), $J(x^k) \to 0$; hence for any $\varepsilon > 0$ there is a finite k with $J(x^k) \le \varepsilon$, so the finite-stopping version terminates after finitely many steps with an ε-feasible point.

4.6. Auxiliary lemmas

We begin with two preparatory results used repeatedly below.

Lemma 4.2 (Nonexpansiveness of the projection)**.** *Let $K \subseteq H$ be nonempty closed convex. For every $y \in H$ and $z \in K$,*

$$\|P_K(y) - z\|^2 \le \|y - z\|^2 - \|y - P_K(y)\|^2 \le \|y - z\|^2 . \tag{19}$$

Proof. Put $p := P_K(y)$. By Proposition 3.2, $\langle y - p,\, z - p\rangle \le 0$, so expanding $\|y - z\|^2 = \|y - p\|^2 + 2\langle y - p,\, p - z\rangle + \|p - z\|^2$ and using $\langle y - p,\, p - z\rangle \ge 0$ gives $\|y - z\|^2 \ge \|y - p\|^2 + \|p - z\|^2$, which is (19). □

Lemma 4.3 (Convergence of a nonnegative sequence with summable perturbation)**.** *Let $\{a_k\} \subset [0, \infty)$ and $\{\varepsilon_k\} \subset [0, \infty)$ satisfy $a_{k+1} \le a_k + \varepsilon_k$ for all k and $\sum_{k=0}^{\infty} \varepsilon_k < \infty$. Then $\{a_k\}$ converges to a finite limit.*

Proof. $b_k := a_k - \sum_{l<k} \varepsilon_l$ is nonincreasing (from $a_{k+1} \le a_k + \varepsilon_k$) and bounded below by $-\sum_l \varepsilon_l > -\infty$, hence converges; since $\sum_{l<k} \varepsilon_l \to S < \infty$, $a_k = b_k + \sum_{l<k} \varepsilon_l$ converges. □

On the convex track, convexity gives a quantitative link between the gradient and the violation.

Lemma 4.4 (Gradient inequality, convex track)**.** *For $x^\star \in \Omega$, $i \in I^{\mathrm{cvx}}$, and $x \in K$,*

$$v_i(x)\ \langle \nabla F_i(x),\, x - x^\star\rangle \ge v_i(x)^2.$$

Proof. If $v_i(x) = 0$ both sides vanish. Let $v_i(x) > 0$, i.e. $F_i(x) > b_i$. Define $\varphi : [0, 1] \to \mathbb{R}$ by $\varphi(t) := F_i((1 - t)x + tx^\star)$; since K is convex, $[x, x^\star] \subset K$, so φ is well defined, and by (A1)(b) it is convex, while by (A2) it is differentiable with $\varphi'(0^+) = \langle \nabla F_i(x),\, x^\star - x\rangle$. The one-dimensional gradient inequality gives $\varphi(1) \ge \varphi(0) + \varphi'(0^+)$, i.e. $F_i(x^\star) \ge F_i(x) + \langle \nabla F_i(x),\, x^\star - x\rangle$. As $x^\star \in \Omega$ we have $F_i(x^\star) \le b_i < F_i(x)$, whence $\langle \nabla F_i(x),\, x - x^\star\rangle \ge F_i(x) - b_i = v_i(x)$. Multiplying by $v_i(x) > 0$ gives the claim. □

Remark 1 (Aggregated convex bound)**.** Summing Lemma 4.4 over the approximate active set K_k^{cvx} yields $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \ge \sum_{i \in K_k^{\mathrm{cvx}}} v_i(x^k)^2 = 2\Phi_k^{\mathrm{cvx}}$, a bound used in the proofs of Lemmas 4.8 and 4.10.

On the pseudoconvex track, only a sign is available.

Lemma 4.5 (Sign inequality, pseudoconvex track)**.** *For $x^\star \in \Omega$, $j \in I^{\mathrm{ps}}$, and $x \in K$, $v_j(x)\ \langle \nabla F_j(x),\, x - x^\star\rangle \ge 0$, with strict inequality when $v_j(x) > 0$.*

Proof. If $v_j(x) = 0$ both sides vanish. If $v_j(x) > 0$ then $F_j(x) > b_j \ge F_j(x^\star)$. By (A1)(b), $F_j|_K$ is pseudoconvex; applying the contrapositive (7) with $y = x^\star$ gives $\langle \nabla F_j(x),\, x^\star - x\rangle < 0$, i.e. $\langle \nabla F_j(x),\, x - x^\star\rangle > 0$. Multiplying by $v_j(x) > 0$ yields the strict inequality. □

Finally, two compactness-based norm bounds on the search directions.

Lemma 4.6 (Upper bound for $\|g^{\mathrm{cvx}}(x)\|^2$ via Φ^{cvx})**.** *Assume $I^{\mathrm{cvx}} \neq \emptyset$ and let $B \subset H$ be compact. With $L_B := \max_{i \in I^{\mathrm{cvx}}} \sup_{x \in B} \|\nabla F_i(x)\|$ (finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{cvx}}(x)\|^2 \leq 2\ |I^{\mathrm{cvx}}|\ L_B^2\, \Phi^{\mathrm{cvx}}(x).$$

Proof. Put $K' = K^{\mathrm{cvx}}(x)$, $m' = |K'| \leq |I^{\mathrm{cvx}}|$. By the triangle and Cauchy–Schwarz inequalities,

$$\|g^{\mathrm{cvx}}(x)\| = \Big\| \sum_{i \in K'} v_i(x) \nabla F_i(x) \Big\| \leq \Big(\sum_{i \in K'} v_i(x)^2 \Big)^{1/2} \Big(\sum_{i \in K'} \|\nabla F_i(x)\|^2 \Big)^{1/2}.$$

Since $\sum_{i \in K'} v_i(x)^2 = 2\Phi^{\mathrm{cvx}}(x)$ and $\sum_{i \in K'} \|\nabla F_i(x)\|^2 \leq m' L_B^2 \leq |I^{\mathrm{cvx}}|\, L_B^2$, squaring gives the claim. □

Lemma 4.7 (Upper bound for $\|g^{\mathrm{ps}}(x)\|$)**.** *Assume $I^{\mathrm{ps}} \neq \emptyset$ and let $B \subset H$ be compact. With $L'_B := \max_{j \in I^{\mathrm{ps}}} \sup_{x \in B} \|\nabla F_j(x)\|$ and $V_B := \max_{j \in I^{\mathrm{ps}}} \sup_{x \in B} v_j(x)$ (both finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{ps}}(x)\| \leq |I^{\mathrm{ps}}|\ V_B\, L'_B.$$

Proof. By the triangle inequality, $\|g^{\mathrm{ps}}(x)\| = \big\|\sum_{j \in K^{\mathrm{ps}}(x)} v_j(x) \nabla F_j(x)\big\| \leq \sum_{j \in K^{\mathrm{ps}}(x)} v_j(x)\, \|\nabla F_j(x)\| \leq |K^{\mathrm{ps}}(x)|\, V_B L'_B \leq |I^{\mathrm{ps}}|\, V_B L'_B$. □

4.7. Boundedness and decay of the iteration error

The central result is a novel one-step quasi-Fejér inequality (conceptually rooted in the framework of Combettes 10), from which we extract boundedness of $\{x_k\}$ and decay of the convex-track violation. Set

$$c_0 := \frac{\rho_{\mathrm{cvx}}(2 - \rho_{\mathrm{cvx}})}{2} \in \Big(0, \tfrac{1}{2}\Big], \qquad C_0 := \frac{2 + \rho_{\mathrm{cvx}}}{2 - \rho_{\mathrm{cvx}}} \in (1, \infty), \tag{20}$$

and

$$T_k := \begin{cases} \dfrac{(2\Phi_k^{\mathrm{cvx}})^2}{\big\|g_k^{\mathrm{cvx}}\big\|^2} & \text{if } \|g_k^{\mathrm{cvx}}\| > 0, \\ 0 & \text{if } \|g_k^{\mathrm{cvx}}\| = 0. \end{cases} \tag{21}$$

For fixed $x^\star \in \Omega$ write $R_k^{x^\star} := \big\langle g_k^{\mathrm{ps}},\, x^k - x^\star \big\rangle$.

Lemma 4.8 (One-step quasi-Fejér inequality)**.** *For every $x^\star \in \Omega$ and $k \geq 0$,*

$$\Big\|x^{k+1} - x^\star\Big\|^2 \leq \Big\|x^k - x^\star\Big\|^2 - c_0 \mu_k T_k - 2(1 - \mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} + C_0 \gamma_k^2. \tag{22}$$

Proof. Apply Lemma 4.2 with $y = y^k$ and $z = x^\star \in K$ and use $y^k = x^k - d_k$ from (18):

$$\Big\|x^{k+1} - x^\star\Big\|^2 \leq \Big\|x^k - x^\star\Big\|^2 - 2\Big\langle d_k,\, x^k - x^\star \Big\rangle + \|d_k\|^2 . \tag{23}$$

Cross term. Expanding d_k,

$$\left\langle d_k,\, x^k - x^\star \right\rangle = \mu_k \eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle + (1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star}. \tag{24}$$

If $\|g_k^{\mathrm{cvx}}\| = 0$ then $\eta_k^{\mathrm{cvx}} = 0$ and the first term vanishes. If $\|g_k^{\mathrm{cvx}}\| > 0$, Remark 1 gives $\left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle \geq 2\Phi_k^{\mathrm{cvx}}$; multiplying by $\mu_k \eta_k^{\mathrm{cvx}} \geq 0$ and substituting $\eta_k^{\mathrm{cvx}} = 2\rho_{\mathrm{cvx}} \Phi_k^{\mathrm{cvx}} / \|g_k^{\mathrm{cvx}}\|^2$,

$$\mu_k \eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star \right\rangle \geq \mu_k \rho_{\mathrm{cvx}} \frac{(2\Phi_k^{\mathrm{cvx}})^2}{\left\|g_k^{\mathrm{cvx}}\right\|^2} = \mu_k \rho_{\mathrm{cvx}} T_k. \tag{25}$$

Quadratic term. By Young's inequality $\|u+v\|^2 \leq (1+\epsilon)\|u\|^2 + (1+\epsilon^{-1})\|v\|^2$ with $u = \mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}}$, $v = (1-\mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}$,

$$\|d_k\|^2 \leq (1+\epsilon)\rho_{\mathrm{cvx}}^2 \mu_k^2 T_k + (1+\epsilon^{-1})(1-\mu_k)^2 (\eta_k^{\mathrm{ps}})^2 \left\|g_k^{\mathrm{ps}}\right\|^2.$$

By definition $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\}$, so $\eta_k^{\mathrm{ps}} \|g_k^{\mathrm{ps}}\| \leq \gamma_k$ and hence $(\eta_k^{\mathrm{ps}})^2 \|g_k^{\mathrm{ps}}\|^2 \leq \gamma_k^2$; with $(1-\mu_k)^2 \leq 1$,

$$\|d_k\|^2 \leq (1+\epsilon)\rho_{\mathrm{cvx}}^2 \mu_k^2 T_k + (1+\epsilon^{-1})\gamma_k^2. \tag{26}$$

Choose $\epsilon := (2-\rho_{\mathrm{cvx}})/(2\rho_{\mathrm{cvx}}) > 0$. Then $(1+\epsilon)\rho_{\mathrm{cvx}} = (2+\rho_{\mathrm{cvx}})/2$, so $(1+\epsilon)\rho_{\mathrm{cvx}}^2 = \rho_{\mathrm{cvx}}(2+\rho_{\mathrm{cvx}})/2$, and $1+\epsilon^{-1} = (2+\rho_{\mathrm{cvx}})/(2-\rho_{\mathrm{cvx}}) = C_0$. Combining (23), (24), (25) and (26),

$$\left\|x^{k+1} - x^\star\right\|^2 \leq \left\|x^k - x^\star\right\|^2 + \left[-2\mu_k\rho_{\mathrm{cvx}} + (1+\epsilon)\rho_{\mathrm{cvx}}^2\mu_k^2\right] T_k - 2(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} + C_0\gamma_k^2.$$

The bracket equals $\mu_k\rho_{\mathrm{cvx}}\big(-2 + \frac{2+\rho_{\mathrm{cvx}}}{2}\mu_k\big)$. Since $\rho_{\mathrm{cvx}} < 2$, for all $\mu_k \in [0,1]$ the factor $-2 + \frac{2+\rho_{\mathrm{cvx}}}{2}\mu_k$ is at most $-(2-\rho_{\mathrm{cvx}})/2 < 0$ (its maximum over $[0,1]$ is at $\mu_k = 1$); multiplying by $\mu_k\rho_{\mathrm{cvx}} \geq 0$ gives the bracket $\leq -c_0\mu_k$. This yields (22). □

Lemma 4.9 (Boundedness and finite series)**.** *The sequence $\{x^k\}$ is bounded. Moreover, for each fixed $x^\star \in \Omega$:* (i) $\{\|x^k - x^\star\|^2\}$ *converges in* $[0,\infty)$*;* (ii) $\sum_{k=0}^{\infty} \mu_k T_k < \infty$*;* (iii) $\sum_{k=0}^{\infty}(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} < \infty$.

Proof. Fix $x^\star \in \Omega$ and set $a_k := \|x^k - x^\star\|^2$. By Lemma 4.5, $R_k^{x^\star} \geq 0$, and $\mu_k, T_k, (1-\mu_k)\eta_k^{\mathrm{ps}} \geq 0$, so (22) gives $a_{k+1} \leq a_k + C_0\gamma_k^2$. Since $\sum_k C_0\gamma_k^2 < \infty$ by (A3)(iii), Lemma 4.3 yields (i), and boundedness of $\{x^k\}$ follows. Summing (22) from 0 to $N-1$,

$$a_N + \sum_{k=0}^{N-1}\big(c_0\mu_k T_k + 2(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star}\big) \leq a_0 + C_0 \sum_{k=0}^{N-1}\gamma_k^2.$$

The right side is bounded uniformly in N; letting $N \to \infty$ and using $c_0 > 0$ with nonnegative terms gives (ii) and (iii). □

Lemma 4.10 (Convex-track violation vanishes)**.** $\lim_{k\to\infty} J_k^{\mathrm{cvx}} = 0$.

Proof. If $I^{\mathrm{cvx}} = \emptyset$ then $J_k^{\mathrm{cvx}} = 0$ for all k. Assume $I^{\mathrm{cvx}} \neq \emptyset$ and put $B := \overline{\mathrm{conv}}\{x^k : k \geq 0\}$, compact by Lemma 4.9. By (A2) and compactness,

$$L := \max_{i \in I^{\mathrm{cvx}}} \sup_{x \in B} \|\nabla F_i(x)\| < \infty, \qquad V := \max_{i \in I} \sup_{x \in B} |F_i(x) - b_i| < \infty.$$

If $V = 0$ then all $v_i(x^k) = 0$ and $J_k^{\mathrm{cvx}} = 0$; assume $V > 0$. If $L = 0$ then each F_i, $i \in I^{\mathrm{cvx}}$, is constant on B, so $g_k^{\mathrm{cvx}} = 0$ and, by Remark 1, $0 = \langle g_k^{\mathrm{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\mathrm{cvx}} \geq (J_k^{\mathrm{cvx}})^2$, giving $J_k^{\mathrm{cvx}} = 0$; assume $L > 0$. Set $m^{\mathrm{cvx}} := |I^{\mathrm{cvx}}| \geq 1$.

Consider a step with $J_k^{\mathrm{cvx}} > 0$ and pick $i^\star \in I^{\mathrm{cvx}}$ with $v_{i^\star}(x^k) = J_k^{\mathrm{cvx}}$. Since $v_{i^\star}(x^k) \geq J_k^{\mathrm{cvx}} - \delta$, we have $i^\star \in K_k^{\mathrm{cvx}}$, whence

$$\Phi_k^{\mathrm{cvx}} \geq \tfrac{1}{2} v_{i^\star}(x^k)^2 = \tfrac{1}{2}(J_k^{\mathrm{cvx}})^2 > 0. \tag{27}$$

Fix $x^\star \in \Omega$. Remark 1 gives $\langle g_k^{\mathrm{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\mathrm{cvx}} > 0$, so $g_k^{\mathrm{cvx}} \neq 0$; with Lemma 4.6, $\|g_k^{\mathrm{cvx}}\|^2 \leq 2m^{\mathrm{cvx}} L^2 \Phi_k^{\mathrm{cvx}}$, and with (27),

$$T_k = \frac{(2\Phi_k^{\mathrm{cvx}})^2}{\|g_k^{\mathrm{cvx}}\|^2} \geq \frac{2\Phi_k^{\mathrm{cvx}}}{m^{\mathrm{cvx}} L^2} \geq \frac{(J_k^{\mathrm{cvx}})^2}{m^{\mathrm{cvx}} L^2}. \tag{28}$$

For a lower bound on μ_k: for all $i \in I$ and $x^k \in B$, $v_i(x^k) \leq |F_i(x^k) - b_i| \leq V$, so $J_k^{\mathrm{cvx}} \leq V$ and $J_k^{\mathrm{ps}} \leq V$; when $J_k^{\mathrm{cvx}} > 0$,

$$\mu_k = \frac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}} \geq \frac{J_k^{\mathrm{cvx}}}{2V}. \tag{29}$$

Combining (28) and (29), for $J_k^{\mathrm{cvx}} > 0$, $\mu_k T_k \geq (J_k^{\mathrm{cvx}})^3/(2Vm^{\mathrm{cvx}}L^2)$; this also holds (as $0 \geq 0$) when $J_k^{\mathrm{cvx}} = 0$. By Lemma 4.9(ii),

$$\sum_{k=0}^{\infty} (J_k^{\mathrm{cvx}})^3 \leq 2Vm^{\mathrm{cvx}} L^2 \sum_{k=0}^{\infty} \mu_k T_k < \infty, \tag{30}$$

so $(J_k^{\mathrm{cvx}})^3 \to 0$ and thus $J_k^{\mathrm{cvx}} \to 0$. □

Proposition 4.11 (Asymptotic regularity)**.** $\sum_{k=0}^{\infty} \left\|x^{k+1} - x^k\right\|^2 < \infty$*; in particular* $\left\|x^{k+1} - x^k\right\| \to 0$.

Proof. Since $x^k \in K$ and $x^{k+1} = P_K(y^k)$, Lemma 4.2 with $y = y^k$, $z = x^k$ gives $\left\|x^{k+1} - x^k\right\| \leq \left\|y^k - x^k\right\| = \|d_k\|$. By (26) and $\mu_k^2 \leq \mu_k$, $\|d_k\|^2 \leq (1+\epsilon)\rho_{\mathrm{cvx}}^2 \mu_k T_k + C_0 \gamma_k^2$. Summing and using Lemma 4.9(ii) and (A3)(iii), both series are finite. □

4.8. Strong convergence to a feasible point

The goal of the analysis is strong convergence of $\{x^k\}$ to a feasible point. The argument has three stages: exploit convex-track feasibility (Lemma 4.10) to confine the cluster set to the convex half; use a contradiction argument with pseudoconvexity to produce

a cluster point in Ω; and combine with distance convergence to upgrade to convergence of the whole sequence.

Theorem 4.12 (Strong convergence)**.** *Under Assumption 1, the sequence $\{x^k\}$ generated by Algorithm 1 converges strongly to a point $x^\infty \in \Omega$.*

Proof. By Lemma 4.9, $\{x^k\}$ is bounded; let $C := \{\bar{x} \in H : \exists\, k_l \to \infty,\ x^{k_l} \to \bar{x}\}$ be its (nonempty, compact) cluster set. Since K is closed and $x^k \in K$, $C \subseteq K$.

Stage 1 (convex half). Fix $\bar{x} \in C$ and $x^{k_l} \to \bar{x}$. By (A2) each F_i is continuous, so $v_i(x^{k_l}) \to v_i(\bar{x})$; for $i \in I^{\mathrm{cvx}}$, $v_i(x^{k_l}) \le J^{\mathrm{cvx}}_{k_l} \to 0$ by Lemma 4.10, so by squeezing $v_i(\bar{x}) = 0$, i.e. $F_i(\bar{x}) \le b_i$. Hence

$$C \subseteq \bigcap_{i \in I^{\mathrm{cvx}}} \{x \in H : F_i(x) \le b_i\}. \tag{31}$$

If $I^{\mathrm{ps}} = \emptyset$, then (31) gives $C \subseteq K \cap \bigcap_{i \in I^{\mathrm{cvx}}} \{F_i \le b_i\} = \Omega$, so $C \cap \Omega = C \neq \emptyset$. Assume henceforth $I^{\mathrm{ps}} \neq \emptyset$.

Stage 2 (pseudoconvex half by contradiction). Suppose, for contradiction, that $C \cap \Omega = \emptyset$. By (31) and the partition $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$, for each $\bar{x} \in C$ there is $j \in I^{\mathrm{ps}}$ with $F_j(\bar{x}) > b_j$, i.e. $J^{\mathrm{ps}}(\bar{x}) > 0$. As a maximum of finitely many continuous functions, J^{ps} is continuous and positive on the compact set C, so $\alpha := \min_{x \in C} J^{\mathrm{ps}}(x) > 0$.

Claim 1 (uniform lower bound for J^{ps}_k). There exist $\tau_0 > 0$ and k_0 with $J^{\mathrm{ps}}_k \ge \tau_0$ for all $k \ge k_0$. Indeed, put $\tau_0 := \alpha/2$. If the claim failed there would be $k_l \to \infty$ with $J^{\mathrm{ps}}_{k_l} < \tau_0$; extracting (same notation) $x^{k_l} \to \tilde{x} \in C$, continuity gives $J^{\mathrm{ps}}(\tilde{x}) \le \tau_0 < \alpha$, contradicting the definition of α.

Claim 2 ($\mu_k \to 0$). For $k \ge k_0$, $\mu_k = J^{\mathrm{cvx}}_k/(J^{\mathrm{cvx}}_k + J^{\mathrm{ps}}_k) \le J^{\mathrm{cvx}}_k/\tau_0 \to 0$ by Lemma 4.10. Hence there is $k_1 \ge k_0$ with $1 - \mu_k \ge 1/2$ for $k \ge k_1$.

Claim 3 (uniform lower bound for $R^{x^\star}_k$). Fix $x^\star \in \Omega$. There exist $\xi_0 > 0$ and $k_2 \ge k_1$ with $R^{x^\star}_k \ge \xi_0$ for all $k \ge k_2$. Suppose not: there is $k_l \to \infty$ with $R^{x^\star}_{k_l} \to 0$. As $\{x^{k_l}\}$ is bounded, extract $x^{k_l} \to \bar{x} \in C$; since I^{ps} has finitely many subsets, extract further so that $K^{\mathrm{ps}}_{k_l} \equiv K^\star$ is constant. By assumption $J^{\mathrm{ps}}(\bar{x}) > 0$; choose $j^\star \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(\bar{x})$, so $v_{j^\star}(\bar{x}) = J^{\mathrm{ps}}(\bar{x}) > 0$. By continuity $v_{j^\star}(x^{k_l}) \to v_{j^\star}(\bar{x})$ and $J^{\mathrm{ps}}_{k_l} \to J^{\mathrm{ps}}(\bar{x})$, so $J^{\mathrm{ps}}_{k_l} - v_{j^\star}(x^{k_l}) \to 0$; with $\delta > 0$ from (A3)(i), for large l we get $v_{j^\star}(x^{k_l}) \ge J^{\mathrm{ps}}_{k_l} - \delta$, hence $j^\star \in K^{\mathrm{ps}}_{k_l} = K^\star$. Each summand of $R^{x^\star}_{k_l} = \sum_{j \in K^\star} v_j(x^{k_l}) \left\langle \nabla F_j(x^{k_l}),\, x^{k_l} - x^\star \right\rangle$ is nonnegative by Lemma 4.5, so keeping only $j^\star$,

$$R^{x^\star}_{k_l} \ge v_{j^\star}(x^{k_l}) \left\langle \nabla F_{j^\star}(x^{k_l}),\, x^{k_l} - x^\star \right\rangle.$$

Letting $l \to \infty$ and using continuity of $v_{j^\star}$ and $\nabla F_{j^\star}$,

$$\liminf_{l \to \infty} R^{x^\star}_{k_l} \ge v_{j^\star}(\bar{x}) \left\langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \right\rangle =: \Lambda.$$

Since $j^\star \in I^{\mathrm{ps}}$ and $x^\star \in \Omega$, $F_{j^\star}(x^\star) \le b_{j^\star} < F_{j^\star}(\bar{x})$; the contrapositive (7) with $x = \bar{x}$, $y = x^\star$ gives $\langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle > 0$, and with $v_{j^\star}(\bar{x}) > 0$ we get $\Lambda > 0$, contradicting $R^{x^\star}_{k_l} \to 0$. This proves Claim 3.

Reaching the contradiction. By Lemma 4.7 on $B := \overline{\mathrm{conv}}\{x^k\}$, $\left\|g^{\mathrm{ps}}_k\right\| \le |I^{\mathrm{ps}}|\, V_B L'$ with $L' := \max_{j \in I^{\mathrm{ps}}} \sup_B \|\nabla F_j\| < \infty$. Put $G' := \max\{1, |I^{\mathrm{ps}}|\, V_B L'\}$; then $\max\{1, \left\|g^{\mathrm{ps}}_k\right\|\} \le G'$ for all k, so $\eta^{\mathrm{ps}}_k = \gamma_k / \max\{1, \left\|g^{\mathrm{ps}}_k\right\|\} \ge \gamma_k/G'$. For $k \ge k_2$,

combining with $1-\mu_k \geq 1/2$ and $R_k^{x^\star} \geq \xi_0$,

$$\sum_{k=0}^{\infty}(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} \geq \sum_{k\geq k_2}\frac{1}{2}\cdot\frac{\gamma_k}{G'}\cdot\xi_0 = \frac{\xi_0}{2G'}\sum_{k\geq k_2}\gamma_k = +\infty$$

by (A3)(iii), contradicting Lemma 4.9(iii). Hence $C\cap\Omega\neq\emptyset$.

Stage 3 (whole-sequence convergence). Fix $\bar{x}\in C\cap\Omega$. By Lemma 4.9(i) with $x^\star=\bar{x}$, $\left\|x^k-\bar{x}\right\|^2 \to \ell \geq 0$. Since $\bar{x}\in C$, some subsequence $x^{k_l}\to\bar{x}$, so $\left\|x^{k_l}-\bar{x}\right\|^2\to 0$, forcing $\ell=0$; thus $x^k\to\bar{x}$. Set $x^\infty:=\bar{x}\in\Omega$. □

Remark 2 (Role of δ)**.** The parameter $\delta>0$ is not necessary for convergence. When $\delta=0$ one argues by subsequence: along $x^{k_l}\to\bar{x}\in C$, choose $j_l\in\arg\max_{j\in I^{\mathrm{ps}}} v_j(x^{k_l})$; since I^{ps} is finite, extract $j_l\equiv j^\star$, and $v_{j^\star}(x^{k_l})=J^{\mathrm{ps}}(x^{k_l})\to J^{\mathrm{ps}}(\bar{x})>0$, so $j^\star$ attains the maximal violation at $\bar{x}$ and the pseudoconvexity argument goes through. The role of $\delta>0$ is to simplify: a maximizer at $\bar{x}$ automatically lies in $K_{k_l}^{\mathrm{ps}}$ for large l, avoiding a 'finitely many indices' extraction. Larger δ enlarges $K_k^{\mathrm{cvx}}, K_k^{\mathrm{ps}}$, which aids numerical stability when several $v_i(x^k)$ are nearly equal.

Remark 3 (A best-violation rate)**.** From (30), $\sum_k (J_k^{\mathrm{cvx}})^3 < \infty$, whence $\min_{k\leq N} J_k^{\mathrm{cvx}} = O(N^{-1/3})$. This bounds the *best* convex violation over the first N steps, not J_N^{cvx} itself; under (A1)–(A3) alone we assert only qualitative convergence.

4.9. Degenerate constraint structures

When the constraint structure degenerates (only a convex track or only a pseudoconvex track), DTAP and its analysis simplify, yielding two corollaries of Theorem 4.12.

Corollary 4.13 (Purely convex constraints)**.** *If $I^{\mathrm{ps}}=\emptyset$, then $J_k^{\mathrm{ps}}\equiv 0$, $g_k^{\mathrm{ps}}\equiv 0$, and $\mu_k\equiv 1$ whenever $J_k^{\mathrm{cvx}}>0$, so Algorithm 1 reduces to $x^{k+1}=P_K(x^k-\eta_k^{\mathrm{cvx}}g_k^{\mathrm{cvx}})$. The sequence converges strongly to a point $x^\infty\in\Omega=K\cap\bigcap_{i\in I^{\mathrm{cvx}}}\{F_i\leq b_i\}$.*

Proof. Here $d_k=\eta_k^{\mathrm{cvx}}g_k^{\mathrm{cvx}}$, so no Young splitting is needed and (23) reads $\left\|x^{k+1}-x^\star\right\|^2 \leq \left\|x^k-x^\star\right\|^2 - 2\eta_k^{\mathrm{cvx}}\left\langle g_k^{\mathrm{cvx}}, x^k-x^\star\right\rangle + (\eta_k^{\mathrm{cvx}})^2\left\|g_k^{\mathrm{cvx}}\right\|^2$. If $\|g_k^{\mathrm{cvx}}\|=0$ then $\eta_k^{\mathrm{cvx}}=0$, $T_k=0$, and the bound is $\left\|x^{k+1}-x^\star\right\|^2\leq\left\|x^k-x^\star\right\|^2$. Otherwise Lemma 4.4 gives $\left\langle g_k^{\mathrm{cvx}}, x^k-x^\star\right\rangle \geq 2\Phi_k^{\mathrm{cvx}}$, and substituting $\eta_k^{\mathrm{cvx}} = 2\rho_{\mathrm{cvx}}\Phi_k^{\mathrm{cvx}}/\left\|g_k^{\mathrm{cvx}}\right\|^2$ together with $(\eta_k^{\mathrm{cvx}})^2\left\|g_k^{\mathrm{cvx}}\right\|^2 = \eta_k^{\mathrm{cvx}}\cdot 2\rho_{\mathrm{cvx}}\Phi_k^{\mathrm{cvx}}$ yields, after simplification, $2\eta_k^{\mathrm{cvx}}\left\langle g_k^{\mathrm{cvx}}, x^k-x^\star\right\rangle - (\eta_k^{\mathrm{cvx}})^2\left\|g_k^{\mathrm{cvx}}\right\|^2 \geq \rho_{\mathrm{cvx}}(2-\rho_{\mathrm{cvx}})(2\Phi_k^{\mathrm{cvx}})^2/\left\|g_k^{\mathrm{cvx}}\right\|^2 = 2c_0T_k$. Hence

$$\left\|x^{k+1}-x^\star\right\|^2 \leq \left\|x^k-x^\star\right\|^2 - 2c_0T_k, \tag{32}$$

valid for all k. Thus $\{\left\|x^k-x^\star\right\|^2\}$ is nonincreasing and convergent, and summing gives $\sum_k T_k<\infty$. As in Lemma 4.10, when $J_k^{\mathrm{cvx}}>0$ one has $T_k\geq (J_k^{\mathrm{cvx}})^2/(m^{\mathrm{cvx}}L^2)$ (note the exponent 2 here, versus 3 in (30), since no μ_k factor is present), so $\sum_k (J_k^{\mathrm{cvx}})^2<\infty$ and $J_k^{\mathrm{cvx}}\to 0$. Finally, $I^{\mathrm{ps}}=\emptyset$ makes Stage 2 of Theorem 4.12 vacuous (indeed $C\subseteq\Omega$ already from (31)), and Stage 3 gives $x^k\to x^\infty\in\Omega$. □

Corollary 4.14 (Purely pseudoconvex constraints)**.** *If $I^{\mathrm{cvx}} = \emptyset$, then $J_k^{\mathrm{cvx}} \equiv 0$, $g_k^{\mathrm{cvx}} \equiv 0$, and $\mu_k \equiv 0$ whenever $J_k^{\mathrm{ps}} > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}})$ with $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\}$. The sequence converges strongly to a point $x^\infty \in \Omega$.*

Proof. Now $d_k = \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}$ with $\|d_k\|^2 \le \gamma_k^2$, so (23) gives $\|x^{k+1} - x^\star\|^2 \le \|x^k - x^\star\|^2 - 2\eta_k^{\mathrm{ps}} R_k^{x^\star} + \gamma_k^2$, which is (22) in this degenerate case ($T_k = 0$, $1 - \mu_k = 1$). Since $R_k^{x^\star} \ge 0$ (Lemma 4.5), $a_{k+1} \le a_k + \gamma_k^2$, and Lemma 4.3 gives convergence of $\{\|x^k - x^\star\|^2\}$ and boundedness; summation gives $\sum_k \eta_k^{\mathrm{ps}} R_k^{x^\star} < \infty$. Stage 1 of Theorem 4.12 gives $C \subseteq K$ (the convex part being vacuous); Stage 2 with $\mu_k = 0$ (so $1 - \mu_k = 1$) produces the same contradiction $\sum \eta_k^{\mathrm{ps}} R_k^{x^\star} = +\infty$, so $C \cap \Omega \neq \emptyset$; Stage 3 gives $x^k \to x^\infty \in \Omega$. □

5. A unified feature-selection framework

We apply the normal-box NSFP to feature selection. We build on two supervised models that have already been tested in practice. The first is the fractional-programming model of Wang et al. [32], which is solved by a projection neural network. The second is the fuzzy-rough-set model AFRSFCV of Wang et al. [33], which is solved by unconstrained softmax optimization. We add *no* new constraint of our own. Instead, we take the three constraint functions directly from the two models' objectives, namely two convex terms from the fuzzy-rough model and one pseudoconvex ratio from the fractional model. Together they form a single normal-box NSFP that satisfies Assumption 1, so DTAP can solve it. Throughout, $U = \{x_1, \dots, x_n\}$ is the sample set, $A = \{a_1, \dots, a_p\}$ the features, $X \in \mathbb{R}^{n \times p}$ the data matrix (column k being feature a_k), $y \in \{1, \dots, r\}^n$ the class labels, and the weight vector w lives on the standard simplex $K := \Delta_{p-1} = \{w \in \mathbb{R}^p : w_k \ge 0,\ \mathbf{1}_p^\top w = 1\}$, a compact convex set that serves as the base set. Sections 5.1 and 5.2 recall the two source models and the quantities they produce. Section 5.3 then shows that these quantities give the three constraints of a single normal-box NSFP that DTAP can solve. Section 5.4 fixes the selected subset from the resulting weights. Finally, Section 5.5 adds a sparse-projection variant that enforces an explicit feature budget.

5.1. The fractional-programming model

5.1.1. Input quantities

Definition 5.1 (Supervised similarity matrix)**.** The *supervised similarity matrix* $S \in \mathbb{R}^{p \times p}$ has entries

$$s_{ij} = \max\Big\{0,\ \frac{I(F_i; F_j; y)}{H(F_i) + H(F_j)}\Big\}, \qquad i, j = 1, \dots, p, \tag{33}$$

where $I(F_i; F_j; y) = I(F_i; y) + I(F_j; y) - I(F_i, F_j; y)$ is the multi-information, $H(\cdot)$ is entropy, and all quantities are estimated from equal-width histograms with K_{bin} bins per feature [4].

Definition 5.2 (Redundancy matrix)**.** The *redundancy matrix* $Q \in \mathbb{R}^{p \times p}$ is obtained by a diagonal shift

$$Q := \delta_Q I_p + S, \qquad \delta_Q \ge -\min\{0, \lambda_{\min}(S)\}, \tag{34}$$

which guarantees $Q \succeq 0$.

The condition $Q \succeq 0$ makes the quadratic form $w^\top Q w$ convex on $\mathbb{R}^p$.

Definition 5.3 (Fisher-score vector)**.** The *Fisher-score vector* $\rho \in \mathbb{R}^p$ has components

$$\rho_k := \frac{\sum_{t=1}^{r} n_t(\mu_{kt} - \mu_k)^2}{\sum_{t=1}^{r} n_t\, \sigma_{kt}^2 + \varepsilon_\rho}, \qquad k = 1, \ldots, p, \tag{35}$$

where n_t is the size of class t, μ_{kt}, σ_{kt}^2 are the class-t mean and variance of a_k, μ_k is the overall mean, and $\varepsilon_\rho > 0$ avoids division by zero.

5.1.2. The N-Fscore baseline

Following [32], supervised selection is the fractional program

$$\min_{w \in K} \frac{w^\top Q w}{\rho^\top w}, \qquad K = \Delta_{p-1}, \tag{36}$$

Its numerator measures redundancy and its denominator relevance, so making the ratio small lowers redundancy while keeping relevance high. The ratio is pseudoconvex on K whenever $\rho_k > 0$ for all k [32]. Wang et al. solve it with a projection neural network (PNN) coupled to a k-winners-take-all stage. Our baseline *N-Fscore* runs only the PNN stage with ρ set to the Fisher score of (35) and passes the resulting weights to the shared selection rule of Section 5.4, so it differs from the other methods only in how $w^\star$ is computed. The PNN dynamics and their explicit-Euler integration are given in Appendix A.

5.2. The fuzzy-rough-set model

Rough-set feature selection uses the uncertainty in the data to score features. In the survey of Liu et al. [24], these methods are organized along a model axis (classical, neighbourhood, fuzzy) and a measure axis (approximation, information, granularity). Fuzzy rough sets work on real-valued data directly, without discretization, and allow membership in $[0, 1]$, which suits numeric data [20]. The AFRSFCV model of Wang et al. [33] belongs to this family and adds class variance to the fuzzy similarity relation.

5.2.1. Input quantities

Let $D = \{d\}$ be the decision attribute with r classes $D_1, \ldots, D_r$, where $D_t := \{x_i \in U : d(x_i) = t\}$. We write $a_k(x_i)$ for the value of x_i on a_k and $t(x_i)$ for its class index.

Definition 5.4 (Class-variance fuzzy similarity)**.** With $\mu_{kt} := \frac{1}{|D_t|} \sum_{x_i \in D_t} a_k(x_i)$ and $\sigma_{kt}^2 := \frac{1}{|D_t|} \sum_{x_i \in D_t} (a_k(x_i) - \mu_{kt})^2$, the *class-variance fuzzy similarity* of x_i, x_j on a_k is

$$R_k(x_i, x_j) := \exp\Big(-\frac{[a_k(x_i) - a_k(x_j)]^2}{2\, \sigma_{k,t(x_i)}^2}\Big). \tag{37}$$

Definition 5.5 (Fuzzy lower approximation)**.** With the class indicator $D_t(x_j) := 1$ if $x_j \in D_t$ and 0 otherwise, the *fuzzy lower approximation* of D_t via a_k is

$$\underline{R_k}(D_t)(x_i) := \min_{x_j \in U} \max\{1 - R_k(x_i, x_j),\, D_t(x_j)\}, \qquad x_i \in U, \tag{38}$$

collected into $\underline{R_k}(D) \in \mathbb{R}^{n \times r}$ with (i, t) entry $\underline{R_k}(D_t)(x_i) \in [0, 1]$.

This measures the certainty of $x_i \in D_t$ using only a_k. Let $Y \in \{0, 1\}^{n \times r}$ be the one-hot label matrix, $Y_{it} = 1$ iff $x_i \in D_t$.

Definition 5.6 (Decision matrix)**.** The *decision matrix* $P \in \{0, 1\}^{n \times p}$ has

$$P_{ik} := \begin{cases} 1 & \text{if } \arg\max_{1 \le t \le r} \underline{R_k}(D_t)(x_i) = t(x_i), \\ 0 & \text{otherwise,} \end{cases} \tag{39}$$

so $P_{ik} = 1$ iff feature a_k classifies x_i correctly under the maximal-lower-approximation rule.

5.2.2. The AFRSFCV baseline

AFRSFCV [33] uses $\underline{R_k}(D)$, P, and Y, and minimizes, over a softmax parametrization,

$$\min_{\tau \in \mathbb{R}^p} L(\tau) := \frac{1}{2} \Big\| Y - \sum_{k=1}^{p} w_k(\tau) \underline{R_k}(D) \Big\|_F^2 + \frac{\lambda_1}{2} \big\| \mathbf{1}_n - P w(\tau) \big\|^2 + \frac{\lambda_2}{2} \|w(\tau)\|^2, \tag{40}$$

where $\| \cdot \|_F$ is the Frobenius norm and the softmax $w_k(\tau) = e^{\tau_k} / \sum_{l=1}^{p} e^{\tau_l}$ keeps $w(\tau) \in K = \Delta_{p-1}$. The first term measures the gap between the label matrix Y and the lower-approximation combination. The second penalizes wrong-class confidence, where $\sum_k P_{ik} w_k$ is the confidence of a correct call. The third is Tikhonov regularization. Our baseline minimizes $L(\tau)$ by gradient descent for a fixed budget and passes the final weight $w^\star = w(\tau^T) \in K$ to the shared selection rule of Section 5.4. The gradient and update are given in Appendix A.

5.3. Reduction to a nonlinear split feasibility problem

Both models seek a weight vector $w \in \mathbb{R}^p$ on the same feature set, with w_k encoding the relative importance of a_k. The normalization $w_k \ge 0$, $\sum_k w_k = 1$ places w on the standard simplex $K = \Delta_{p-1}$, a compact convex set that serves as the base set of the split feasibility problem. From the two models' objectives we extract three constraint functions.

Definition 5.7 (The three constraint functions)**.** Define

$$F_1(w) := \Big\| Y - \sum_{k=1}^{p} w_k \underline{R_k}(D) \Big\|_F^2, \tag{41}$$

$$F_2(w) := \big\| \mathbf{1}_n - Pw \big\|^2, \tag{42}$$

$$F_3(w) := \frac{w^\top Q w}{\rho^\top w}, \tag{43}$$

where $Y, \underline{R_k}(D), P$ are from Section 5.2, Q, ρ from Section 5.1, and F_3 is defined on $\{w : \rho^\top w > 0\}$.

Here F_1, F_2 are the first two terms of the AFRSFCV loss (40), validated in [33], and F_3 is exactly the fractional objective (36) of [32]; no artificial function is added.

Proposition 5.8 (Constraint regularity)**.** *Under $Q \succeq 0$ and $\rho_k > 0$ for all k: F_1 and F_2 are convex on $\mathbb{R}^p$, and F_3 is pseudoconvex on K. Consequently, with $I^{\mathrm{cvx}} = \{1, 2\}$ and $I^{\mathrm{ps}} = \{3\}$, the feasibility problem*

$$\textit{find } w^\star \in \Omega, \qquad \Omega := \{w \in K : F_i(w) \le b_i,\ i = 1, 2, 3\}, \tag{44}$$

with thresholds $b_1, b_2, b_3 > 0$, is an instance of the normal-box NSFP (8) *satisfying Assumption 1 (provided $\Omega \neq \emptyset$).*

Proof. Each of F_1, F_2 is the squared norm of an affine map of w: indeed $w \mapsto Y - \sum_k w_k \underline{R_k}(D)$ and $w \mapsto \mathbf{1}_n - Pw$ are affine, and the squared norm of an affine map is convex; hence F_1, F_2 are convex on $\mathbb{R}^p$, meeting the convex-track requirement of (A1)(b). For F_3, the numerator $w^\top Q w$ is a positive semidefinite quadratic ($Q \succeq 0$ by (34)) and the denominator $\rho^\top w$ is a positive linear form on K; by [32] such a ratio is pseudoconvex on K, meeting the pseudoconvex-track requirement. All F_i are smooth with continuous gradients, giving (A2); (A1)(a) is the standing nonemptiness of Ω, and (A3) concerns only the algorithm parameters. Thus (44) is of the form (8). □

By Proposition 5.8, DTAP (Algorithm 1) applies directly to (44). Although F_3 is only defined where $\rho^\top w > 0$, when $\rho_k > 0$ for all k we have $\rho^\top w = \sum_k \rho_k w_k \ge \min_k \rho_k > 0$ for every $w \in K$, so K lies in the open half-space and, since every DTAP iterate stays in K, both $F_3(w)$ and $\nabla F_3(w)$ are well defined and continuous along the sequence. As the convergence analysis in Section 4 uses constraint values only on K, Theorem 4.12 applies: the DTAP iterates for (44) converge strongly to a feasible $w^\star \in \Omega$.

Remark 4 (Per-iteration cost of the instance)**.** For the instance (44) with n samples, p features, and r classes, the per-iteration cost of Section 4.5 is dominated by the constraint evaluations: $F_1, \nabla F_1$ from (41) cost $O(nrp)$ (a weighted combination of the p matrices $\underline{R_k}(D)$); $F_2, \nabla F_2$ from (42) cost $O(np)$ through P; $F_3, \nabla F_3$ from (43) cost $O(p^2)$ through the quadratic form $w^\top Q w$; and the simplex projection costs $O(p \log p)$. One DTAP iteration thus runs in $O(nrp + p^2)$ time and $O(nrp)$ memory (the lower-approximation matrices), the same per-iteration order as the AFRSFCV and Hu subgradient baselines, which operate on the same matrices; the difference lies in the *number* of iterations, quantified in Section 6.4 (Table B3).

Problem (44) can also be solved by the subgradient method for *quasi-convex* fea-

sibility of Hu et al. [19]: since convex and pseudoconvex functions are quasi-convex, all three constraints are quasi-convex on K, so (44) is a quasi-convex feasibility problem. We use two variants as baselines, *SG-const* (constant step) and *SG-dyn* (dynamic step), both following the most-violated-constraint control scheme (at tolerance level α) to form a search direction, then updating and projecting onto K. Unlike DTAP, the SG-dyn variant must estimate a Hölder modulus of the constraint functions per dataset, a cost that DTAP avoids through the Polyak step on the convex track.

5.4. From weights to a selected subset

Once a weight vector $w^\star$ is obtained by any method, the selected feature set is fixed by a shared *cumulative-threshold* rule. Sorting the weights in decreasing order $w^\star_{(1)} \geq \cdots \geq w^\star_{(p)}$, take the smallest count s whose cumulative mass reaches a threshold $\theta \in (0,1)$:

$$s := \min\Big\{ s' \in \{1,\dots,p\} : \sum_{j=1}^{s'} w^\star_{(j)} \geq \theta \Big\}, \tag{45}$$

and keep the s features with the largest weights. The same rule is applied to every method, so that differences in results come only from the weight-computation stage.

5.5. Explicit feature budgets via sparse projection

The cumulative-threshold rule (45) selects features by how *concentrated* $w^\star$ is, not by their number. On high-dimensional data this is a limitation: because the convex constraints F_1, F_2 spread weight across many features to reproduce the label structure, $w^\star$ is nearly flat, so reaching cumulative mass θ forces most coordinates to be kept. At $\theta = 0.9$, DTAP retains 33/36 features on Satimage and 49/57 on Spambase (Table 3), contrary to the goal $|A^\star| \ll p$. We remove this limitation by moving the sparsity requirement *inside* the iteration.

Definition 5.9 (s-sparse simplex)**.** For a budget $s \in \{1,\dots,p\}$, the s-sparse simplex is $\Delta^{(s)}_{p-1} := \{w \in \Delta_{p-1} : \|w\|_0 \leq s\}$, where $\|w\|_0 = |\{k : w_k \neq 0\}|$.

Although $\Delta^{(s)}_{p-1}$ is nonconvex, a union of the s-dimensional faces of the simplex, its Euclidean projection has a closed form via the Greedy Selector and Simplex Projector (GSSP) [22]: for $y \in \mathbb{R}^p$, let $S(y)$ index the s largest entries of y; then

$$\big(P_{\Delta^{(s)}_{p-1}}(y)\big)_k = \begin{cases} \big(P_{\Delta_{s-1}}(y_{S(y)})\big)_k & k \in S(y), \\ 0 & \text{otherwise}, \end{cases} \tag{46}$$

that is, keep the s largest coordinates, project them onto the standard simplex, and zero the rest; the cost is $O(p \log p)$, the same order as P_K.

The DTAP-S *algorithm.* Replacing the base projection in Step 2 of Algorithm 1 with $w^{k+1} = P_{\Delta^{(s)}_{p-1}}(y^k)$ yields the sparse variant DTAP-S$_s$. Every iterate then carries at most s nonzeros, so the selected set is exactly $\mathrm{supp}(w^\star)$ with $|A^\star| \leq s$: the budget s is an *explicit, tunable* feature count that replaces the indirect threshold θ.

Remark 5 (A deliberate heuristic trade-off)**.** Since $\Delta_{p-1}^{(s)}$ is nonconvex, the projection (46) is no longer nonexpansive, so the strong-convergence guarantee of Theorem 4.12 does not transfer; DTAP-S is thus a heuristic that trades the convex-case guarantee for genuine feature reduction. Empirically (Section 6.3) the iterates remain stable and the best iterate is retained. The budget s caps the support while the dual-track update is otherwise unchanged, so all the modelling of Section 5.3 is preserved.

In summary, the redundancy and Fisher quantities of the fractional model and the lower-approximation, decision, and label matrices of the fuzzy-rough model jointly furnish the three constraints of a single nonlinear split feasibility problem. The reduction reuses only validated objectives, preserving the meaning of both source models while bringing the problem into a form solvable by DTAP (either exactly, via the strongly convergent base algorithm, or under an explicit budget, via the sparse variant DTAP-S) and remaining open to new selection criteria added as further convex or pseudoconvex constraints.

6. Numerical experiments

We evaluate DTAP on the feature-selection problem (44) against four baselines, namely the two base models N-Fscore [32] and AFRSFCV [33] and the two subgradient variants SG-const and SG-dyn of Hu et al. [19]. The experiments use eight UCI benchmarks that vary in sample size, dimension, class count, and balance. We report selected-set size, classification accuracy, area under the ROC curve, and selection stability, together with runtime and convergence behaviour.

6.1. Experimental setup

6.1.1. Datasets

Eight UCI datasets [21] are used, with n from a few hundred to over ten thousand, p from 7 to 57, and r from 2 to 10 (Table 2).

Table 2. UCI datasets used in the experiments.

Dataset	Samples n	Features p	Classes r	Balance
Glass	214	9	6	Imbalanced
Diabetes	768	8	2	Imbalanced
Segmentation	2 310	18	7	Balanced
Abalone	4 177	8	3	Balanced
Satimage	6 435	36	6	Imbalanced
Yeast	1 484	7	10	Imbalanced
Spambase	4 601	57	2	Imbalanced
Pendigits	10 992	16	10	Balanced

6.1.2. Parameters and shared inputs

On each training split, features are min–max scaled to $[0, 1]$, and the inputs shared by all five methods are computed once. These are the fuzzy lower-approximation matrices $\underline{R_k}(D)$, the label matrix Y, and the decision matrix P (Section 5.2), together with the redundancy matrix Q and the Fisher vector ρ (Section 5.1); multi-information and entropy are estimated from equal-width histograms with 10 bins per feature [4]. The DTAP parameters are fixed across all datasets. We start from $w^0 = (1/p, \ldots, 1/p)^\top$ and use the Polyak parameter $\rho_{\text{cvx}} = 1.98$, the pseudoconvex step $\gamma_k = 1/(k + 0.5)^{0.55}$ (so $\sum_k \gamma_k = \infty$ and $\sum_k \gamma_k^2 < \infty$), the approximate-active parameter $\delta = 10^{-3}$, the tolerance $\varepsilon = 10^{-6}$, and at most 1 000 iterations. The Hu variants use the same 1 000-iteration cap and the most-violated-constraint control scheme, where SG-const sets $v = 10^{-3}$ and SG-dyn estimates a per-dataset Hölder modulus by Dirichlet sampling on the simplex. N-Fscore uses $\varepsilon_{\text{PNN}} = 10^{-3}$, $T = 0.2$, and $N_{\text{step}} = 1\,000$ Euler steps under a fixed budget. AFRSFCV uses learning rate $\eta = 10^{-3}$, weights $\lambda_1 = 1$, $\lambda_2 = 0.5$, and $T = 1\,000$ gradient steps. The shared cumulative threshold is $\theta = 0.9$. The three DTAP parameters have transparent roles. The Polyak relaxation $\rho_{\text{cvx}} \in (0, 2)$ trades per-step decrease against the perturbation constant (we use 1.98 for fast early progress), the exponent $s \in (1/2, 1]$ controls the safeguarded step ($s = 0.55$), and δ only stabilizes the search directions when several violations are nearly equal (Remark 2). These settings are structural rather than tuned per dataset, so one configuration serves the whole suite. Since all five methods consume the same precomputed inputs, any difference in the outputs arises only in the weight-computation stage and the shared threshold rule (45). Further implementation and reproducibility details are given in Appendix B.

6.1.3. Data-driven output thresholds

The thresholds b_1, b_2, b_3 in Ω (44) are set by a data-driven calibration. For each dataset we draw 3 000 points $w^{(s)} \sim \text{Dir}(\mathbf{1}_p)$ on the simplex K and set b_i to a per-dataset percentile of the sampled values $F_i(w^{(s)})$, tuned so that $\Omega \neq \emptyset$ yet is tight enough for the reduction to be effective. Since F_1, F_2 are sums of squared residuals over n samples, b_1, b_2 grow with sample size, whereas b_3, a ratio, stays small. The full procedure and the resulting thresholds are given in Appendix B (Table B1).

6.1.4. Evaluation protocol

Each dataset is split 80/20 in a stratified manner into a development pool and a held-out test set. On the pool we run stratified K-fold cross-validation with $K = 10$, reduced to the size of the minority class when needed ($K = 7$ on Glass, $K = 4$ on Yeast). In each fold, after scaling and input computation, the five methods are run independently to obtain weight vectors, and rule (45) with $\theta = 0.9$ fixes the selected set; a feature is kept if selected in more than half the folds (majority vote). Selection stability across folds is measured by the Nogueira index $\Phi \in [-1, 1]$ [28], with $\Phi = 1$ for perfectly stable selection. The voted feature set then trains two classifiers on the full pool, which are evaluated on the held-out test set. These are an RBF-kernel support vector machine (RBF-SVM, $C = 1$) [12] and a 5-nearest-neighbour classifier (5-NN) [13]. The two classification metrics are accuracy and macro one-vs-rest AUC. For reference, both classifiers are also evaluated on the full feature set.

Discrimination is measured by the macro one-vs-rest AUC, $\frac{1}{r}\sum_c \text{AUC}_c$ [17], and a selection is conventionally called stable when $\Phi \gtrsim 0.75$. Across the eight datasets we compare methods by two-sided Wilcoxon signed-rank and Friedman tests [14] at

$\alpha = 0.05$, with Holm–Bonferroni correction for the family-wise error.

6.2. Classification results

We analyse selection and classification along four axes, namely selected-set size, accuracy, AUC, and selection stability. As the statistical tests at the end of this subsection show, DTAP and the two Hu variants give almost identical results, with Wilcoxon $p = 1$ on three of the four metrics and only a small difference on RBF-SVM AUC; the tables below therefore list DTAP as the representative of the split-feasibility group, with SG-const and SG-dyn compared separately in Table 5 and in Section 6.4.

Table 3. Number of selected features $|A^\star|$ and selection stability (Nogueira index) across folds. Column p is the original feature count. The highest stability among the three selection methods is in bold.

Dataset	p	Selected $\lvert A^\star \rvert$			Nogueira stability		
		N-Fscore	AFRSFCV	DTAP	N-Fscore	AFRSFCV	DTAP
Glass	9	7	7	7	**0.8229**	0.6230	0.6458
Diabetes	8	7	3	7	0.2698	**0.8050**	0.6989
Segmentation	18	8	2	16	0.8904	**1.0000**	0.8094
Abalone	8	4	2	5	0.9485	**1.0000**	**1.0000**
Satimage	36	10	3	33	**0.8809**	0.7083	0.5950
Yeast	7	6	5	6	**1.0000**	**1.0000**	0.7870
Spambase	57	26	4	49	**0.9176**	0.8950	0.8461
Pendigits	16	13	12	15	0.8600	0.1001	**1.0000**

Table 3 shows that DTAP retains a moderate number of features. It retains 16/18 on Segmentation, 33/36 on Satimage, and 49/57 on Spambase, whereas AFRSFCV's softmax retains very few features (2, 3, and 4) and N-Fscore lies in between. High stability alone does not imply quality. AFRSFCV is perfectly stable on several sets, yet its accuracy is low because its selections are overly sparse, as shown next.

Table 4. Classification accuracy (%) and macro one-vs-rest AUC on the held-out test set, averaged over the eight datasets. Per-dataset results are given in Appendix B (Table B2). The best among the three selection methods (N-Fscore, AFRSFCV, DTAP) is in bold.

Method	Accuracy (%)		AUC	
	RBF-SVM	5-NN	RBF-SVM	5-NN
Full	80.61	78.71	0.9164	0.8870
N-Fscore	77.05	77.12	0.8994	0.8770
AFRSFCV	75.34	75.16	0.8807	0.8759
DTAP	**80.17**	**78.63**	**0.9136**	**0.8938**

Table 4 reports the averages over the eight datasets, and Table B2 in Appendix B gives the per-dataset results. On average DTAP remains within half an accuracy point of the full feature set, is the best of the three selection methods on all four metrics, and even exceeds the full set on the 5-NN AUC. Per dataset, it matches the full set with RBF-SVM on Segmentation, Satimage, and Diabetes, and exceeds it with 5-NN on Segmentation and Pendigits. Against the base models the advantage is largest on many-class or high-dimensional sets. On Segmentation DTAP exceeds N-Fscore by 12.77 accuracy points and AFRSFCV by 14.50, and on Satimage by 2.02 and 8.70.

AFRSFCV's gaps stem from its overly sparse selections, which discard features needed for multi-class discrimination, whereas the convex constraints $F_1 \leq b_1, F_2 \leq b_2$ keep DTAP's weights distributed across enough features to preserve the classification structure. The AUC results show the same trend. The largest gap is the SVM-AUC on Diabetes, where AFRSFCV reaches only 0.6617 against DTAP's 0.8098, a direct result of retaining only three features.

Table 5. Two-sided Wilcoxon signed-rank p-values (DTAP versus each method, over 8 datasets) and Friedman p-values for all five methods.

Metric	Wilcoxon (DTAP vs)				Friedman
	N-Fscore	AFRSFCV	SG-const	SG-dyn	
Accuracy, RBF-SVM	0.031	0.016	1.000	1.000	0.001
Accuracy, 5-NN	0.375	0.031	1.000	1.000	0.006
AUC, RBF-SVM	0.016	0.016	0.031	0.938	0.005
AUC, 5-NN	0.078	0.156	1.000	1.000	0.043

Table 5 gives two observations. First, DTAP and the two variants SG-const and SG-dyn are statistically indistinguishable on three of the four metrics: the Wilcoxon p-values equal 1.00 for accuracy on both classifiers and for 5-NN AUC. Their voted selections are in fact identical on seven of the eight datasets, differing only on Spambase (49 features for DTAP against 48 for the SG variants). The tests separate them only on RBF-SVM AUC, where the macro-AUC differs by small margins even when the selection is identical, giving $p = 0.031$ against SG-const and $p = 0.938$ against SG-dyn. This near-equivalence justifies using DTAP as their representative. Second, DTAP outperforms both base models on the SVM metrics. Against N-Fscore, $p = 0.031$ for accuracy and $p = 0.016$ for AUC; against AFRSFCV, $p = 0.016$ for both accuracy and AUC. After Holm–Bonferroni correction over the four comparisons these SVM advantages become borderline (adjusted $p \approx 0.06$), which reflects the small sample of eight datasets, but the Friedman test still rejects equality of the five methods on SVM accuracy ($p = 0.001$), SVM-AUC ($p = 0.005$), 5-NN accuracy ($p = 0.006$), and 5-NN AUC ($p = 0.043$). Overall, DTAP's advantage over the base models is concentrated on the SVM classifier and is weaker for 5-NN.

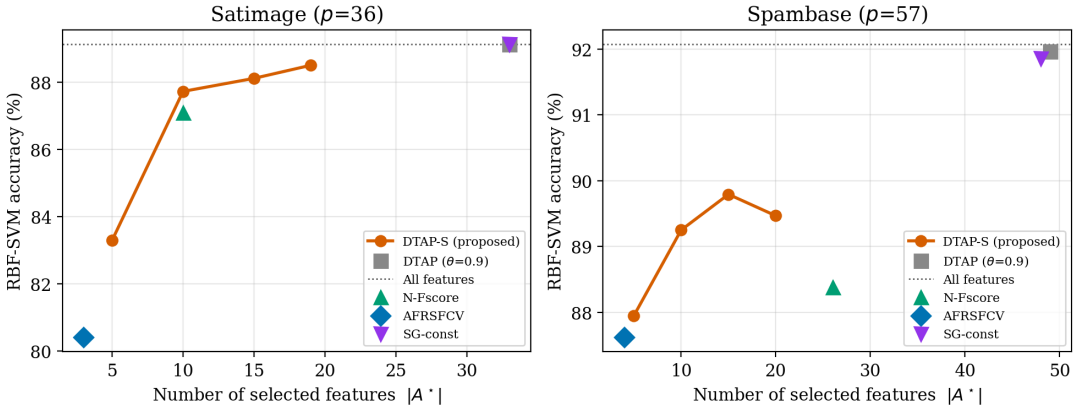
6.3. Explicit feature budgets with **dtap-*S***

Section 5.5 observed that the cumulative rule lets DTAP (and the SG variants, which share the rule) keep most features on high-dimensional data. We now test the sparse variant DTAP-S on the two highest-dimensional benchmarks, Satimage (p=36) and Spambase (p=57), under the identical 80/20 split and majority-vote-over-ten-folds protocol, sweeping the budget s. Table 6 and Fig. 1 report the outcome.

Table 6. Sparse selection with DTAP-S on the two highest-dimensional benchmarks, reporting selected size $|A^\star|$, held-out accuracy (%), and Nogueira stability Φ. The budget s in DTAP-S$_s$ is an explicit cap on $|A^\star|$. Best accuracy at a matched small budget is in bold.

Dataset	Method	$\lvert A^\star\rvert$	RBF-SVM	5-NN	Φ
Satimage (p=36)	Full	36	89.12	91.14	–
	N-Fscore	10	87.10	87.80	0.881
	AFRSFCV	3	80.42	80.96	0.708
	DTAP	33	89.12	91.22	0.595
	DTAP-S$_{15}$	15	88.11	89.98	0.852
	DTAP-S$_{10}$	10	**87.72**	**89.43**	0.852
Spambase (p=57)	Full	57	92.07	90.34	–
	N-Fscore	26	88.38	85.78	0.918
	AFRSFCV	4	87.62	88.49	0.895
	DTAP	49	91.97	89.25	0.846
	DTAP-S$_{10}$	10	**89.25**	**88.71**	1.000
	DTAP-S$_5$	5	87.95	88.71	1.000

Figure 1. RBF-SVM accuracy versus selected size $|A^\star|$ on Satimage and Spambase. At similar selected sizes the DTAP-S curve (orange) lies above the baseline points, and on Spambase DTAP-S$_{10}$ outperforms N-Fscore with fewer features and higher accuracy.



On Satimage and Spambase, where a small set of useful features exists, DTAP-S makes DTAP select a genuinely small set and outperforms the baselines at similar budgets. On Spambase, DTAP-S$_{10}$ uses only 10 features yet reaches 89.25/88.71 (SVM/5-NN) with perfect stability $\Phi = 1.000$, while N-Fscore needs 26 features for the lower 88.38/85.78 at $\Phi = 0.918$. DTAP-S$_5$ with 5 features likewise outperforms AFRSFCV with 4 features on all three measures. On Satimage at a matched 10 features, DTAP-S$_{10}$ (87.72/89.43) again exceeds N-Fscore (87.10/87.80). The budget also cuts DTAP's own selection from 33/49 features to 10 while *raising* stability from 0.595/0.846 to 0.852/1.000, since a fixed budget forces a consistent set across folds (Fig. 1).

6.4. Runtime and convergence

We now compare the computational cost and convergence behaviour of the five methods. Table 7 reports the wall-clock time, iteration count, and per-iteration cost of each

method, averaged first over the cross-validation folds and then over the eight datasets; the per-dataset values are given in Appendix B (Table B3).

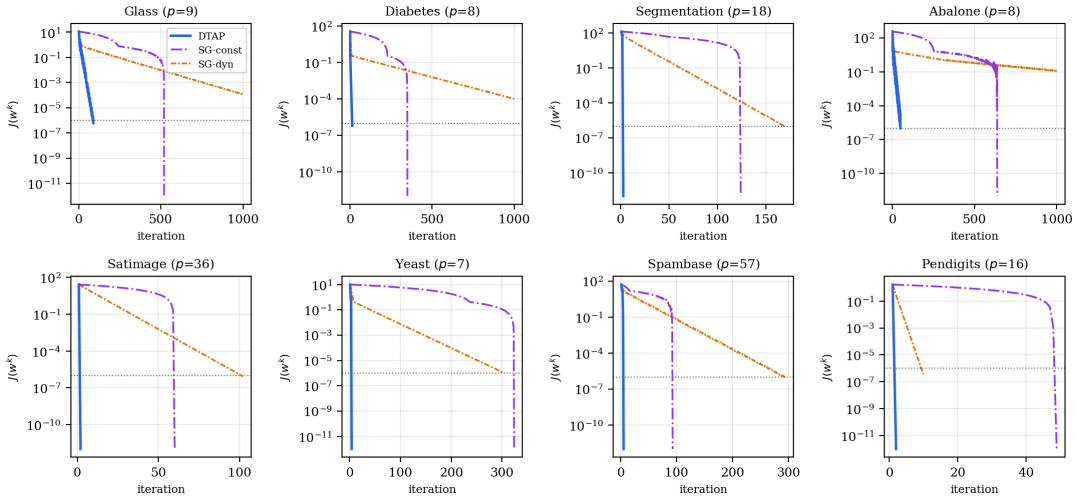
Table 7. Wall-clock time, iteration count, and per-iteration cost, averaged over the eight datasets (fold averages per dataset). The time averages are dominated by the largest datasets; per-dataset values are in Appendix B (Table B3). Best per column in bold.

Method	Time (s)	Iterations	Cost (ms/iter)
N-Fscore	0.057	999.0	**0.057**
AFRSFCV	1.620	1000.0	1.620
SG-const	0.180	300.7	1.758
SG-dyn	0.883	484.8	25.569
DTAP	**0.007**	**25.1**	0.935

Table 7 shows that DTAP has the lowest runtime, and the per-dataset values in Table B3 confirm this on every one of the eight datasets. The advantage does not come from the per-iteration cost. As noted in Remark 4, DTAP, AFRSFCV, and SG-const share the same $O(nrp + p^2)$ per-iteration order, since all three compute the constraint gradients on the same input matrices. On Satimage, for instance, they measure 2.587, 4.140, and 4.492 ms per iteration. The advantage comes from the iteration count. Thanks to the stopping test $J(w^k) \leq \varepsilon$, DTAP stops after very few iterations, whereas AFRSFCV uses the full 1 000-iteration budget and SG-dyn often reaches it too. The gap is largest on the large sets. On Pendigits DTAP needs 0.004 s versus 4.893 s for AFRSFCV and 1.761 s for SG-dyn, and on Satimage the three are 0.005, 4.140, and 2.032 s. N-Fscore runs 999 low-cost Euler steps that do not recompute the residual on large matrices, but it is still slower than DTAP on all eight sets.

A separate convergence experiment on the full data of each set gives a consistent picture (Fig. 2). DTAP and SG-const drive J below ε on all eight datasets, whereas SG-dyn reaches feasibility on only five. The reason is that the estimated Hölder modulus is often very large, about 6.5×10^3 on Segmentation for instance, which makes the dynamic step tiny and convergence slow. This agrees with Theorem 4.12 on the strong convergence of DTAP and shows the benefit of the Polyak step on the convex track. DTAP avoids estimating the Hölder modulus and still reaches feasibility in few iterations.

Figure 2. Violation $J(w^k)$ (log scale) versus iteration for DTAP, SG-const, and SG-dyn on the eight datasets. DTAP (solid) reaches feasibility in very few iterations on every dataset, while SG-dyn (dashed) stalls on several because the estimated Hölder modulus makes its step size too small.



6.5. Further analyses of robustness, fidelity, and ablation

Five additional experiments examine the reasons behind the results above; we summarise them briefly. *(i) Constant-free robustness.* Every subgradient baseline must fix a step constant, namely v for SG-const and the Hölder modulus L for SG-dyn, and it reaches feasibility only inside a narrow band of that constant. Averaged over five datasets and nine swept values, SG-const succeeds for 6.2/9 values and SG-dyn for 3.8/9, whereas DTAP carries no such constant and reaches feasibility from the single standard start on all eight datasets. *(ii) Deterministic convergence.* From the standard start DTAP reaches $J \leq \varepsilon$ on all eight sets, SG-const does so as well but with 5–81× more iterations, and SG-dyn succeeds on only five because it stalls when the estimated modulus shrinks its step. The DTAP trajectory also has no run-to-run variance. *(iii) Performance profile.* On iteration count DTAP is fastest on all eight datasets ($\rho(1) = 1.0$), which confirms that the speed-up of Section 6.4 comes from the iteration count, not the per-iteration cost. *(iv) Selection fidelity on synthetic ground truth.* On controlled data (n=600, p=30, four classes, eight informative features), DTAP recovers all informative features at every redundancy level, rejects the most noise, and reaches the highest precision of the three selection methods, whereas the N-Fscore PNN under-selects and discards 12–44% of the informative features. *(v) Dual-track ablation.* Collapsing DTAP to a single track supports the design. A single diminishing-step subgradient on all constraints fails to reach feasibility within budget on four of five datasets, and a Polyak-only variant has no guarantee once the pseudoconvex constraint binds; only the full dual-track scheme is both guaranteed and fast. A Nemenyi critical-difference test on SVM accuracy ranks DTAP first (mean rank 2.17), significantly ahead of AFRSFCV.

6.6. Summary of findings

The numerical study supports four conclusions. (i) *Quality.* DTAP matches or exceeds the full-feature accuracy and AUC on most datasets and outperforms both base models on the SVM metrics (Tables 4 and 5). (ii) *Selection size.* The two convex constraints retain enough features to preserve the multi-class structure, in contrast to the overly sparse softmax selections of AFRSFCV (Table 3). (iii) *Speed.* DTAP is the fastest method on all eight datasets, by up to three orders of magnitude on the large sets, and the gain comes from the iteration count rather than the per-iteration cost (Table 7). (iv) *Robust convergence.* DTAP reaches feasibility on all eight datasets, whereas SG-dyn does so on only five because its Hölder-modulus estimate makes the step size too small (Fig. 2). Finally, DTAP and the Hu variants give nearly identical selections (Wilcoxon $p = 1$ on three of the four metrics, with only a small difference on RBF-SVM AUC), so the split-feasibility formulation delivers the quality, while the DTAP design delivers the speed.

7. Conclusion

We studied a nonlinear split feasibility problem with a normal-box output set and mixed convex–pseudoconvex component maps. A Tchebyshev scalarization reduced feasibility to a closed-form max-type violation function, and we proposed DTAP, a dual-track algorithm that takes a residual-adaptive step on the convex constraints and a safeguarded diminishing step on the pseudoconvex ones, fuses the two by a soft weight, and projects only onto the base set. On the convex track the step is the constant-free rule of López et al. [25] extended to the nonlinear setting, licensed now by convexity of the individual components rather than by firm nonexpansiveness of a linear composition; the pseudoconvex track absorbs the remaining constraints on the sign that pseudoconvexity still supplies. Its step sizes need no operator-norm, Lipschitz, or Hölder constant, and we proved strong convergence to a feasible point (Theorem 4.12). Two predecessors are closest. Against the nonlinear multiple-sets solver of Li et al. [23], DTAP uses the same search direction but replaces the global, unverifiable hypothesis that the aggregate proximity function be convex (a hypothesis that a single pseudoconvex component invalidates, leaving the backtracking search possibly undefined) by a per-constraint classification and a closed-form step, and it drops the boundedness and global-Lipschitz requirements of that analysis. Against the subgradient method of Yen et al. [36], with which it shares the max-type scalarization and the active-set Clarke direction, DTAP differs in the admissible class (convex-plus-pseudoconvex, which neither contains nor is contained in their quasilinear class) and in driving the convex part by a non-diminishing, residual-scaled step rather than a prescribed diminishing sequence (Table 1).

On the application side, two previously separate feature-selection models reduce to a single normal-box NSFP whose constraints are drawn from their objectives, and the sparse variant DTAP-S adds an explicit feature budget. On eight UCI benchmarks DTAP preserved or improved accuracy and AUC over the base models, with the clearest gains on the SVM metrics, and reached feasibility substantially faster than the same-class subgradient methods, while DTAP-S gave compact, stable selections that outperform the baselines at matched size. This answers the three questions posed in the introduction.

Several limitations should be stated plainly. The theory delivers a feasibility guar-

antee rather than optimality, with a worst-case best-violation rate of $O(N^{-1/3})$ (Remark 3), and sharper rates need extra structure such as an error bound. Performance depends on the threshold calibration of Section 6.1.3, since the percentiles are tuned per dataset and a poor choice can empty Ω; the calibration is applied identically to the whole split-feasibility group, so it cannot favour DTAP over its same-class competitors. The iteration count can grow with the feature dimension, and the $p \gg n$ regime of text or genomic data is untested. Finally, we report two classifiers and two metrics on eight datasets, a modest sample, so we treat marginal tests such as the 5-NN AUC as inconclusive.

Because criteria enter the model as constraints, further objectives such as sparsity surrogates or fairness penalties can be appended as convex or pseudoconvex constraints without changing the solver. An inertial acceleration would control the asymptotic regularity of Proposition 4.11, and the Hilbert-space analysis points to infinite-dimensional extensions and to other applications of nonlinear split feasibility, such as equilibrium models [36] and constrained learning [31].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

The authors received no specific funding for this work.

Declaration of generative AI use

The authors report that generative AI was not used in the research or the preparation of this manuscript.

Data availability

The eight datasets used in this study are publicly available from the UCI Machine Learning Repository [21]. An implementation of DTAP and the baseline solvers is available from the authors on reasonable request.

ORCID

Nguyen Van The *https://orcid.org/0009-0004-6548-420X*
Tran Ngoc Thang *https://orcid.org/0009-0000-9549-5736*
Nguyen Hong Son *https://orcid.org/0009-0006-3106-8296*

References

[1] H. H. Bauschke and J. M. Borwein. On projection algorithms for solving convex feasibility problems. *SIAM Review*, 38(3):367–426, 1996. .

[2] H. H. Bauschke and P. L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. CMS Books in Mathematics. Springer, New York, 2011. .

[3] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011. .

[4] Gavin Brown, Adam Pocock, Ming-Jie Zhao, and Mikel Luján. Conditional likelihood maximization: a unifying framework for information theoretic feature selection. *Journal of Machine Learning Research*, 13:27–66, 2012.

[5] C. Byrne. Iterative oblique projection onto convex sets and the split feasibility problem. *Inverse Problems*, 18(2):441–453, 2002. .

[6] Y. Censor and T. Elfving. A multiprojection algorithm using Bregman projections in a product space. *Numerical Algorithms*, 8:221–239, 1994. .

[7] Y. Censor, T. Bortfeld, B. Martin, and A. Trofimov. A unified approach for inversion problems in intensity-modulated radiation therapy. *Physics in Medicine and Biology*, 51 (10):2353–2365, 2006. .

[8] J. Chen, M. Postolache, and Y. Yao. S-subgradient projection methods with S-subdifferential functions for nonconvex split feasibility problems. *Symmetry*, 11(12):1517, 2019. .

[9] P. L. Combettes and J.-C. Pesquet. Proximal splitting methods in signal processing. In *Fixed-Point Algorithms for Inverse Problems in Science and Engineering*, pages 185–212. Springer, New York, 2011. .

[10] Patrick L. Combettes. Quasi-fejérian analysis of some optimization algorithms. In D. Butnariu, Y. Censor, and S. Reich, editors, *Inherently Parallel Algorithms in Feasibility and Optimization and Their Applications*, volume 8 of *Studies in Computational Mathematics*, pages 115–152. Elsevier, Amsterdam, 2001. .

[11] Laurent Condat. Fast projection onto the simplex and the ℓ_1 ball. *Mathematical Programming*, 158(1):575–585, 2016. .

[12] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20 (3):273–297, 1995. .

[13] Thomas M. Cover and Peter E. Hart. Nearest neighbour pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967. .

[14] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006. URL `https://jmlr.org/papers/v7/demsar06a.html`.

[15] M. Fukushima. A relaxed projection method for variational inequalities. *Mathematical Programming*, 35(1):58–70, 1986. .

[16] Isabelle Guyon and André Elisseeff. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182, 2003.

[17] David J. Hand and Robert J. Till. A simple generalization of the area under the ROC curve for multiple class classification problems. *Machine Learning*, 45(2):171–186, 2001. .

[18] Tuy Hoang. *Convex Analysis and Global Optimization*. Springer International Publishing, Switzerland, 2nd edition, 2016. .

[19] Y. Hu, G. Li, C. K. W. Yu, and T. L. Yip. Quasi-convex feasibility problems: Subgradient methods and convergence rates. *European Journal of Operational Research*, 298(1):45–58, 2022. .

[20] Richard Jensen and Qiang Shen. New approaches to fuzzy-rough feature selection. *IEEE Transactions on Fuzzy Systems*, 17(4):824–838, 2009. .

[21] Markelle Kelly, Rachel Longjohn, and Kolby Nottingham. The UCI machine learning repository. `https://archive.ics.uci.edu`, 2023. Accessed 2026.

[22] Anastasios Kyrillidis, Stephen Becker, Volkan Cevher, and Christoph Koch. Sparse pro-

jections onto the simplex. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28 of *PMLR*, pages 235–243, 2013.
[23] Z. Li, D. Han, and W. Zhang. A self-adaptive projection-type method for nonlinear multiple-sets split feasibility problem. *Inverse Problems in Science and Engineering*, 21 (1):155–170, 2013. .
[24] Keyu Liu, Xibei Yang, Weiping Ding, Hengrong Ju, Tianrui Li, Jie Wang, and Tengyu Yin. A survey on rough feature selection: Recent advances and challenges. *IEEE/CAA Journal of Automatica Sinica*, 13(3):521–542, 2026. .
[25] G. López, V. Martín-Márquez, F. Wang, and H.-K. Xu. Solving the split feasibility problem without prior knowledge of matrix norms. *Inverse Problems*, 28(8):085004, 2012. .
[26] O. L. Mangasarian. Pseudo-convex functions. *SIAM Journal on Control*, 3(2):281–290, 1965. .
[27] O. L. Mangasarian. *Nonlinear Programming*. Classics in Applied Mathematics. SIAM, Philadelphia, 1994. .
[28] Sarah Nogueira, Konstantinos Sechidis, and Gavin Brown. On the stability of feature selection algorithms. *Journal of Machine Learning Research*, 18(174):1–54, 2018.
[29] B. T. Polyak. Minimization of unsmooth functionals. *USSR Computational Mathematics and Mathematical Physics*, 9(3):14–29, 1969. .
[30] Dipti Theng and Kishor K. Bhoyar. Feature selection techniques for machine learning: a survey of more than two decades of research. *Knowledge and Information Systems*, 66: 1575–1637, 2024. .
[31] Tran Anh Tuan, Nguyen Viet Dung, and Tran Ngoc Thang. A hyper-transformer model for controllable pareto front learning with split feasibility constraints. *Neural Networks*, 179:106571, 2024. .
[32] Y. Wang, X. Li, and J. Wang. A neurodynamic optimization approach to supervised feature selection via fractional programming. *Neural Networks*, 136:194–206, 2021. .
[33] C. Wang, Y. Zhang, S. An, and T. Deng. Adaptive feature selection based on fuzzy rough set fusion model with class variance. *Pattern Recognition*, 170:112014, 2026. .
[34] H.-K. Xu. Iterative methods for the split feasibility problem in infinite-dimensional Hilbert spaces. *Inverse Problems*, 26(10):105018, 2010. .
[35] Q. Yang. The relaxed CQ algorithm solving the split feasibility problem. *Inverse Problems*, 20(4):1261–1266, 2004. .
[36] Le Hai Yen, Nguyen Thi Thanh Huyen, and Le Dung Muu. A subgradient algorithm for a class of nonlinear split feasibility problems: application to jointly constrained nash equilibrium models. *Journal of Global Optimization*, 73:849–868, 2019. .

Appendix A. Baseline solvers

This appendix records the solver mechanics of the two base models, deferred from Sections 5.1.2 and 5.2.2 for readability.

N-Fscore: the projection neural network To solve the fractional program (36), Wang et al. [32] proposed a neurodynamic method coupling a projection neural network (PNN) with a k-winners-take-all network. The PNN is the differential system

$$\varepsilon_{\mathrm{PNN}}\frac{du}{dt} = -Pg(u) - (I-P)\big(u - g(u) + \nabla f((I-P)g(u) + \tfrac{1}{p}\mathbf{1}_p)\big) + \tfrac{1}{p}\mathbf{1}_p, \qquad \text{(A1)}$$

with output $w = g(u)$, where $g(u) = \max\{0, u\}$ is the componentwise ReLU, $P = \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top$, ∇f is the gradient of the objective in (36), and $\varepsilon_{\mathrm{PNN}} > 0$ is a time constant.

We integrate (A1) by explicit Euler on $[0, T]$ with N_{step} steps under a fixed budget (no stopping rule), with T, N_{step} large enough for w to stabilize. For a uniform comparison we use only the PNN part to obtain $w^\star$; the k-WTA stage is replaced by the shared cumulative-threshold rule of Section 5.4.

AFRSFCV: gradient and update The softmax loss (40) is minimized over $\tau \in \mathbb{R}^p$ by gradient descent. Its gradient is

$$\nabla_\tau L(\tau) = \big(\operatorname{diag}(w(\tau)) - w(\tau)w(\tau)^\top\big)\nabla_w L(w(\tau)), \tag{A2}$$

with $\nabla_w L = -\mathbf{M}^\top(\operatorname{vec}(Y) - \mathbf{M}w) + \lambda_1(P^\top P w - P^\top \mathbf{1}_n) + \lambda_2 w$, where $\mathbf{M} \in \mathbb{R}^{nr \times p}$ has k-th column $\operatorname{vec}(\underline{R_k}(D))$. The update $\tau^{t+1} = \tau^t - \eta\,\nabla_\tau L(\tau^t)$ runs for a fixed budget of T steps; the final weight $w^\star = w(\tau^T) \in K$ feeds the shared selection rule of Section 5.4.

Appendix B. Experimental details

This appendix collects the threshold calibration, the implementation details, and the per-dataset classification and runtime results deferred from Section 6.

Output-threshold calibration For each dataset we evaluate F_1, F_2, F_3 at the 3 000 sampled points $w^{(s)} \sim \operatorname{Dir}(\mathbf{1}_p)$ and set b_i to the p_i-th percentile of the resulting values, so about $p_i\%$ of simplex points satisfy $F_i \le b_i$ and the percentile controls the size of the feasible set. The percentiles p_i are tuned per dataset (by sample size, dimension, class count, and balance). The resulting b_i, averaged over folds, appear in Table B1.

Table B1. Output thresholds b_1, b_2, b_3 (fold averages) per dataset.

Dataset	b_1	b_2	b_3
Glass	142	18.3	0.0638
Diabetes	549	98.9	0.0443
Segmentation	1 450	466	0.00956
Abalone	2 990	1 080	0.105
Satimage	4 570	3 200	0.0356
Yeast	887	579	0.0524
Spambase	3 250	1 010	0.0800
Pendigits	7 910	7 080	0.0141

Implementation and reproducibility All five methods share one per-fold pipeline and consume identical precomputed inputs $(\underline{R_k}(D), P, Y, Q, \rho)$. Min–max scaling is fit on each fold's training portion only, multi-information uses 10-bin equal-width histograms [4], Q is shifted by $\delta_Q = -\min\{0, \lambda_{\min}(S)\}$ to keep $Q \succeq 0$, the Fisher denominator uses $\varepsilon_\rho = 10^{-8}$, and the simplex projection is the sort-based $O(p \log p)$ algorithm [11]. Random seeds for the CV splits and the Dirichlet sampling are fixed, so the study is reproducible. Reported runtimes are relative cross-method measurements

and are not sensitive to hardware, because the per-iteration cost is dominated by the shared matrix operations.

Per-dataset classification results Table B2 reports the per-dataset accuracy and macro one-vs-rest AUC that underlie the averages of Table 4.

Table B2.: Per-dataset classification accuracy (%) and macro one-vs-rest AUC on the held-out test set for RBF-SVM and 5-NN. The best among the three selection methods (N-Fscore, AFRSFCV, DTAP) is in bold.

Dataset	Method	Accuracy (%)		AUC	
		RBF-SVM	5-NN	RBF-SVM	5-NN
Glass	Full	76.74	72.09	0.9250	0.9097
	N-Fscore	69.77	**74.42**	0.9023	0.8933
	AFRSFCV	**72.09**	69.77	0.9078	**0.9371**
	DTAP	**72.09**	69.77	**0.9144**	**0.9371**
Diabetes	Full	74.68	71.43	0.8089	0.7356
	N-Fscore	71.43	70.78	0.7956	0.7154
	AFRSFCV	65.58	67.53	0.6617	0.7145
	DTAP	**74.68**	**73.38**	**0.8098**	**0.7766**
Segmentation	Full	93.29	94.37	0.9937	0.9880
	N-Fscore	80.52	87.01	0.9712	0.9801
	AFRSFCV	78.79	86.58	0.9668	0.9646
	DTAP	**93.29**	**94.59**	**0.9940**	**0.9897**
Abalone	Full	61.60	59.81	0.8102	0.7742
	N-Fscore	61.36	**61.00**	0.8065	0.7605
	AFRSFCV	62.56	58.73	0.8113	**0.7752**
	DTAP	**62.68**	60.29	**0.8123**	0.7695
Satimage	Full	89.12	91.14	0.9796	0.9792
	N-Fscore	87.10	87.80	0.9755	0.9705
	AFRSFCV	80.42	80.96	0.9611	0.9306
	DTAP	**89.12**	**91.22**	**0.9796**	**0.9766**
Yeast	Full	58.25	51.52	0.8480	0.7667
	N-Fscore	**58.92**	**51.52**	0.8057	**0.7763**
	AFRSFCV	57.91	**51.52**	0.8218	0.7681
	DTAP	58.59	**51.52**	**0.8380**	0.7585
Spambase	Full	92.07	90.34	0.9663	0.9443
	N-Fscore	88.38	85.78	0.9391	0.9214
	AFRSFCV	87.62	88.49	0.9155	0.9206
	DTAP	**91.97**	**89.25**	**0.9615**	**0.9437**
Pendigits	Full	99.09	98.95	0.9995	0.9982
	N-Fscore	**98.95**	98.64	**0.9996**	0.9982
	AFRSFCV	97.77	97.73	**0.9996**	0.9965
	DTAP	**98.95**	**99.05**	0.9995	**0.9987**

Per-dataset runtime Table B3 reports the per-dataset wall-clock time, iteration count, and per-iteration cost that underlie the averages of Table 7.

Table B3.: Wall-clock time, iteration count, and per-iteration cost (fold averages) of the five methods.

Dataset	Method	Time (s)	Iterations	Cost (ms/iter)
Glass	N-Fscore	0.059	999.0	**0.059**
	AFRSFCV	0.160	1000.0	0.160
	SG-const	0.101	503.1	0.201
	SG-dyn	0.265	991.4	0.267
	DTAP	**0.012**	**75.7**	0.148
Diabetes	N-Fscore	0.057	999.0	**0.057**
	AFRSFCV	0.166	1000.0	0.166
	SG-const	0.070	324.7	0.216
	SG-dyn	0.245	923.6	0.272
	DTAP	**0.002**	**11.7**	0.169
Segmentation	N-Fscore	0.056	999.0	**0.056**
	AFRSFCV	0.712	1000.0	0.712
	SG-const	0.098	124.5	0.786
	SG-dyn	0.341	159.7	2.133
	DTAP	**0.003**	**4.9**	0.557
Abalone	N-Fscore	0.055	999.0	**0.055**
	AFRSFCV	0.289	1000.0	0.289
	SG-const	0.284	881.8	0.322
	SG-dyn	0.415	1000.0	0.415
	DTAP	**0.021**	**88.0**	0.243
Satimage	N-Fscore	0.058	999.0	**0.058**
	AFRSFCV	4.140	1000.0	4.140
	SG-const	0.274	61.0	4.492
	SG-dyn	2.032	107.0	18.998
	DTAP	**0.005**	**2.0**	2.587
Yeast	N-Fscore	0.055	999.0	**0.055**
	AFRSFCV	0.240	1000.0	0.240
	SG-const	0.099	365.0	0.271
	SG-dyn	0.180	377.0	0.501
	DTAP	**0.002**	**11.2**	0.174
Spambase	N-Fscore	0.063	999.0	**0.063**
	AFRSFCV	2.357	1000.0	2.357
	SG-const	0.265	97.0	2.730
	SG-dyn	1.827	309.8	5.902
	DTAP	**0.008**	**5.0**	1.513
Pendigits	N-Fscore	0.056	999.0	**0.056**
	AFRSFCV	4.893	1000.0	4.893
	SG-const	0.246	48.7	5.043
	SG-dyn	1.761	10.0	176.065
	DTAP	**0.004**	**2.0**	2.086